7 Tooth Morphology and Pulpal Access Cavities

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CHAPTER OUTLINE

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- Midroot Considerations
- Apical Considerations

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The dental pulp presents with a variety of configurations and shapes throughout the dentition. Knowledge of the complexity of the root canal system is essential to understand the principles and problems encountered in access preparation from a traditional and contemporary standpoint. This chapter describes and illustrates current perspectives on tooth morphology and explains the techniques necessary to achieve direct access to the root canal system.

Initial anatomical evaluation of the pulpal anatomy is achieved with careful evaluation of two or more periapical radiographs, exposed at different horizontal angulations of the x-ray cone, and is essential, along with cone-beam computed tomography (CBCT) scans in some clinical situations (see Chapter 2). The detailed reading and interpreting of each radiograph before and during root canal procedures is necessary because many teeth present with unusual canal

morphology. Unfortunately, the interpretation of traditional radiographs may not always result in the correct morphologic assessment, particularly when only a buccolingual view is taken. Furthermore, the recent development of microcomputed tomography (μCT) scanning of teeth has greatly increased clinical assessment of these complexities and three-dimensional (3D) relationships found in root canal systems. Therefore the use of a vast array of tools, such as magnification and illumination, is essential. This approach is particularly helpful in posterior teeth, where a vast array of anatomical complexities can be observed (Fig. 7.1).

The use of the dental operating microscope (DOM) that provides superior magnification, increased lighting, and enhanced visibility is recommended to determine the location of root canal orifices in the properly prepared coronal access (Fig. 7.2),[92,105] and has been shown to be superior to the use of the naked eye and magnifying loupes for this assessment.[11,96] As with all advances in technology, however, there are varying points of concurrence and disagreement. For example, one group of investigators determined that dental loupes and the DOM were equally effective for locating mesiopalatal canals in maxillary molars,[13] whereas other studies determined that the DOM did not significantly enhance the ability to locate canals.[34]

Components of the Pulp Cavity

The pulp cavity (Fig. 7.3) is divided into two main portions: the pulp chamber, located in (or extending to just below) the anatomic crown of the tooth, and the root canal (or canals), found in the anatomic root. Other notable features are the pulp horns; accessory, lateral, and furcation canals; canal orifices; isthmus; fins; apical deltas; and apical foramina. The outline of the pulp cavity generally corresponds to the external contour of the tooth. However, factors such as physiologic aging, pathosis, trauma, and occlusion all can modify its dimensions through the production of dentin or reparative (irregular secondary, irritational, and tertiary) dentin (Fig. 7.4).[100]

In single-rooted teeth, pulp chamber extends to the most apical portion of the cervical margin of the crown; in double/multirooted teeth, it extends to the floor of chamber located in the coronal third of the root.[2] The pulp horns are important because the pulp in them is often exposed by caries, trauma, or mechanical invasion, which usually necessitates vital pulp or root canal procedures. Also, the pulp horns undergo rapid mineralization, along with reduction of the size and shape of the pulp chamber because of the formation of reparative/irritational dentin over time.

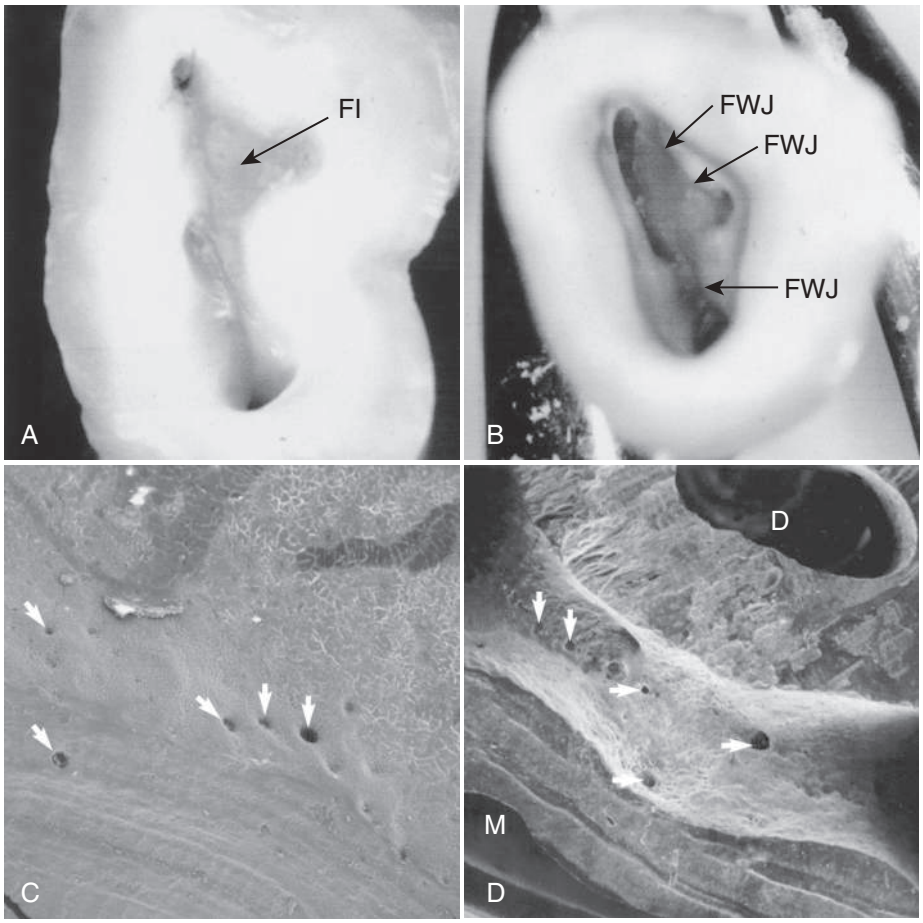


Fig. 7.1 **A,** The color of the pulp chamber floor *(Fl)* is always darker than the walls. The orifices are usually located at the junction of the walls and the floor *(FWJ)*. **B,** The same is true for the maxillary first premolar. **C,** Electron photomicrograph of the pulp chamber floor of a mandibular first molar (×20). **D,** Electron photomicrograph of the furcation surface of a mandibular first molar (×30). *D,* Distal canal; *M,* mesial canals; *arrows*, accessory foramina. (**A** and **B**, From Krasner P, Rankow HJ: Anatomy of the pulp chamber floor, *J Endod* 30:5–16, 2004.)



Fig. 7.2 The dental operating microscope (DOM) has vastly improved locating the position of the coronal canal anatomy.

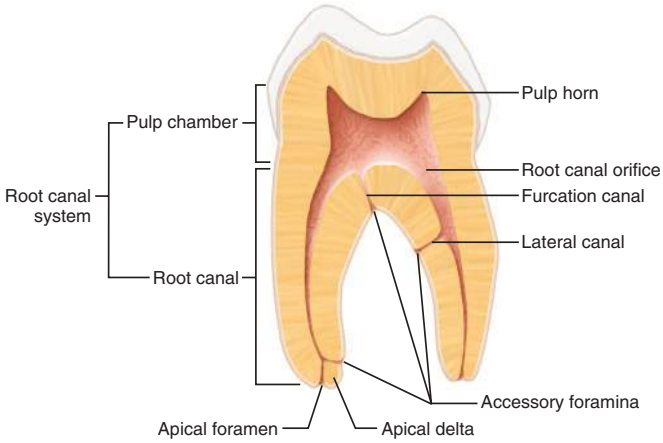


Fig. 7.3 Major anatomic components of the pulp cavity.

The root canal begins as a funnel-shaped canal orifice, generally at or just apical to the cervical line, and ends at the apical foramen (AF), which opens onto the root surface at or within 3 mm of the center of the root apex.[14,35,36,87,113,115] Nearly all root canals are curved, particularly in a faciolingual direction, thereby posing problems during enlargement and shaping procedures because they are not evident on a standard two-dimensional (2D) radiograph. In most cases, the number of root canals corresponds to the number of roots; however, an oval root may have more than one canal.

Accessory canals are minute canals that extend in a horizontal, vertical, or lateral direction from the pulp space to the periodontium. In 74% of cases they are found in the apical third of the root, in 11% in the middle third, and in 15% in the cervical third.[113] The diameter, length, shape, and undulation may vary among accessory canals.[122] Apical deltas are multiple accessory canals that branch out from the main canal at or near the root apex (Figs. 7.5 and 7.6). Accessory canals contain connective tissue and vessels but may not supply the pulp with sufficient circulation to form a collateral source of blood flow. They are formed by the entrapment of periodontal vessels in Hertwig's epithelial root sheath during mineralization.[24] They may play a significant role in the communication of disease processes, serving as avenues for the passage of irritants, primarily from the pulp to the periodontium, although communication of inflammatory processes may occur from either tissue.

Accessory canals that are present in the bifurcation or trifurcation of multirooted teeth are referred to as *furcation canals* (or *chamber canals*) (Fig. 7.7).[37,113,114] These channels form as a result of the entrapment of periodontal vessels during the fusion of the diaphragm, which becomes the pulp chamber floor.[24] In mandibular molars, these canals occur in three distinct patterns (Fig. 7.8). Pulpal inflammation can communicate to the periodontium via these canals and result in furcation lesions in the absence of demonstrable periodontal disease (Fig. 7.9). Likewise, the long-term presence of periodontal furcation lesions may influence the viability of the coronal or radicular pulp tissue when these aberrant channels are present.[39]

Root Canal Anatomy

The significance of canal anatomy has been underscored by studies that demonstrated that the natural variations in canal geometry had a greater effect on the changes that occurred during enlargement and shaping than did the instrumentation techniques used to achieve these objectives.[84–86] Most recently, Versiani and co-investigators have detailed the complete pulpal anatomy of the permanent dentition. In

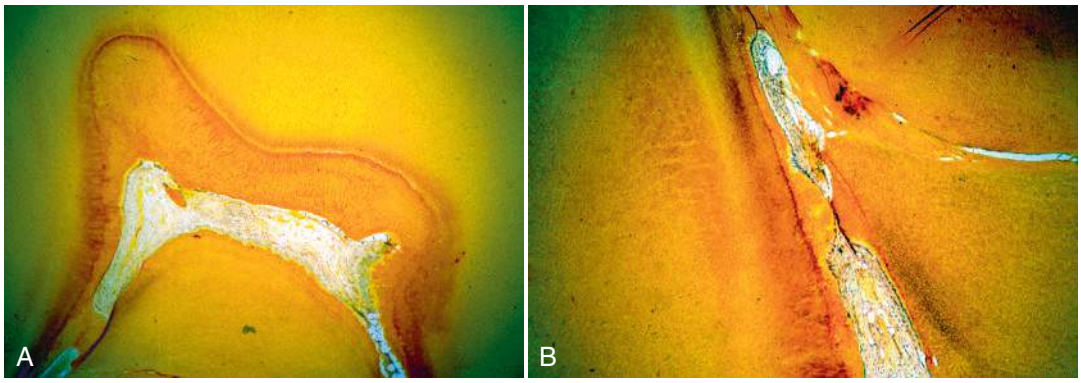


Fig. 7.4 A, Formation of reparative dentin and reduction of the pulp chamber space. **B,** Calcific narrowing and closing of a root canal as it leaves the pulp chamber. (From Gutmann JL, Lovdahl PE: *Problem solving in endodontics*, ed 5, St. Louis, 2011, Elsevier.)



Fig. 7.5 **A,** μCT reconstruction image of a maxillary molar showing the location of accessory canal. **B,** Morphological variations of accessory canal.

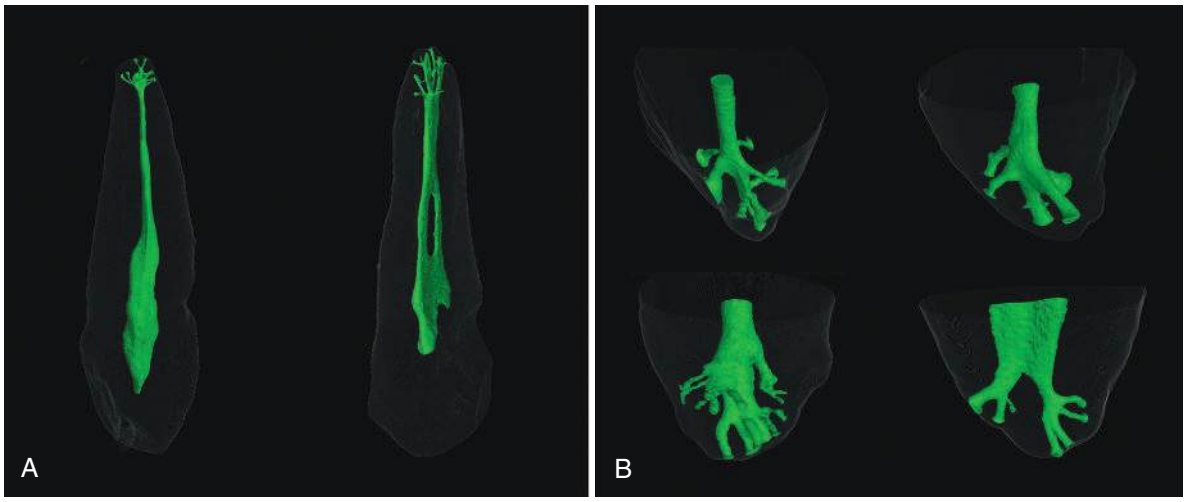


Fig. 7.6 **A,** Apical delta in maxillary canine and maxillary second premolar. **B,** Morphological variations of apical delta.

doing so they have both verified the findings of the last 125 years and have prepared the formative knowledge base for future studies.[110]

Multiple anatomic variations present in teeth are seen in Fig. 7.10. The only tooth that showed all eight possible configurations is the maxillary second premolar.

Similar observations have been described in large population studies with the exceptions that one canal was found in 23% of maxillary laterals, 55% of mesiobuccal roots of maxillary second molars, and 30% of distal roots of mandibular second molars.[17,119] Differences in these studies may be due to variations in the ethnic and racial populations studied. A well-recognized ethnic variant is the higher incidence of single-rooted and C-shaped mandibular second molars in Native American and Asians populations (Figs. 7.11 and 7.12),[26,27,64,65,118] which may

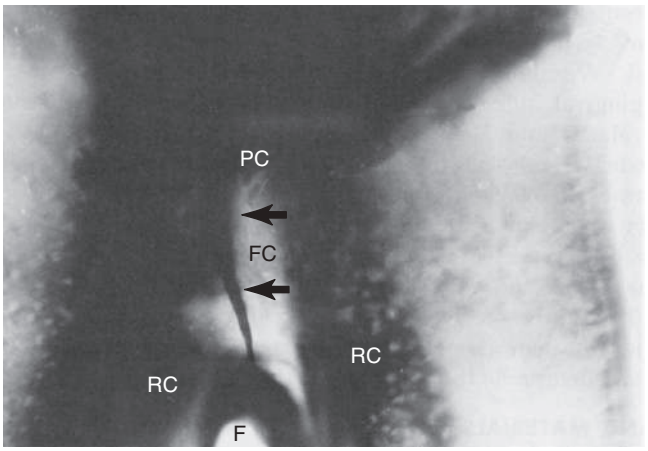


Fig. 7.7 Mandibular first molar showing a furcation canal *(FC, arrows). F,* Furcation; *PC,* pulp chamber floor; *RC,* root canal.

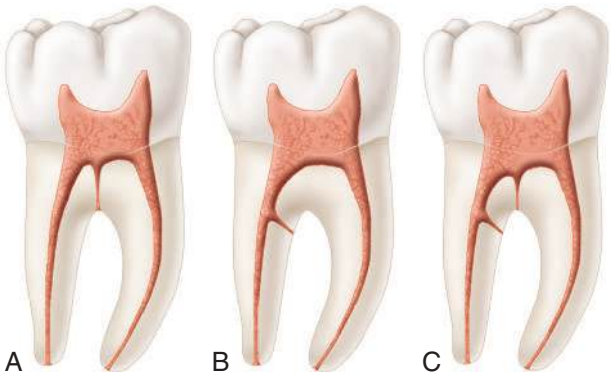


Fig. 7.8 Accessory canals occur in three distinct patterns in the mandibular first molars. **A,** In 13% a single furcation canal extends from the pulp chamber to the intraradicular region. **B,** In 23% a lateral canal extends from the coronal third of a major root canal to the furcation region (80% extend from the distal root canal). **C,** About 10% have both lateral and furcation canals.

also be seen in mandibular premolars and some maxillary molars. Recently, three classification system have been proposed, including (1) root and canal anomalies, (2) root and root canal configurations, and (3) accessory canal morphology that provides more detailed information on the morphological features of teeth and root canals essential for diagnosis and treatment, as well as training and research.[1–3]

CLINICAL DETERMINATION OF THE ROOT CANAL CONFIGURATION

Coronal Considerations

Examination of the pulp chamber floor may reveal clues to the location of orifices and to the type of canal system present. If only one canal is present, it usually is located in the center of the access preparation. If only one orifice is found and it is not in the center of the root, another orifice probably exists on the opposite side (Fig. 7.13). The relationship of the two orifices to each other is also significant (Fig. 7.14). The closer they are, the greater the chance the two canals join at some point in the body of the root. As the distance between orifices in a root increases, the greater is the chance the canals will remain separate. The more separation between orifices, the less the degree of canal curvature.[19]

The direction a file takes when introduced into an orifice is also important. If the first file inserted into the distal canal of a mandibular molar points either in a buccal or lingual direction, a second canal is often present. If two canals are present, they will be smaller than a single canal. (See μCTs and cross sections of all tooth groups later in this chapter.

Midroot Considerations

As the canal leaves the coronal portion of the root and blends into the midroot portion, many changes can occur, including fins, webs, culs-de-sac, and isthmuses (also called



Fig. 7.9 A, Furcation lesion. **B,** Healing occurred after treatment.

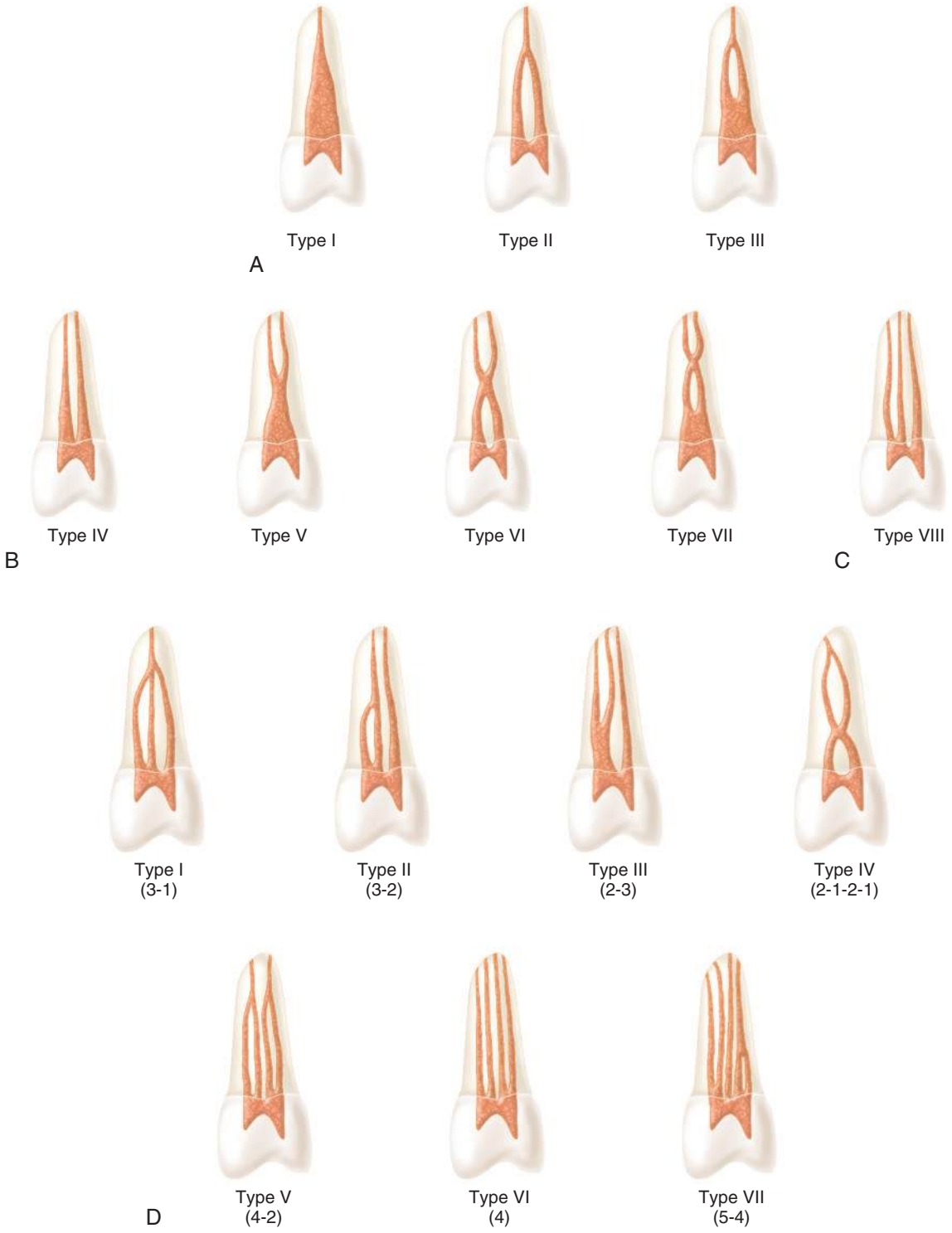


Fig. 7.10 Diagrammatic representation of canal configurations. **A,** One canal at apex based on the work of Vertucci. **B,** Two canals at apex based on the work of Vertucci. **C,** Three canals at apex based on the work of Vertucci. **D,** Supplemental canal configurations based on the work of Gulabivala and colleagues on a Burmese population. (From Vertucci FJ, Seelig A, Gillis R: Root canal morphology of the human maxillary second premolar, *Oral Surg Oral Med Oral Pathol* 38:456–464, 1974.)

anastomoses). These structures are narrow, ribbon-shaped communications between two root canals that contain pulp tissue, or they may midroot portion of the canal. Isthmuses are found in 15% of anterior teeth; in maxillary premolar teeth, they are found in 16% at the 1-mm level from the apex and in 52% at the 6-mm level, which puts them primarily in the middle third of the canal (Fig. 7.15). The prevalence of an isthmus increases in the mesiobuccal root of the maxillary first molar, from 30% to 50% near the junction of middle and apical thirds of the root. Eighty

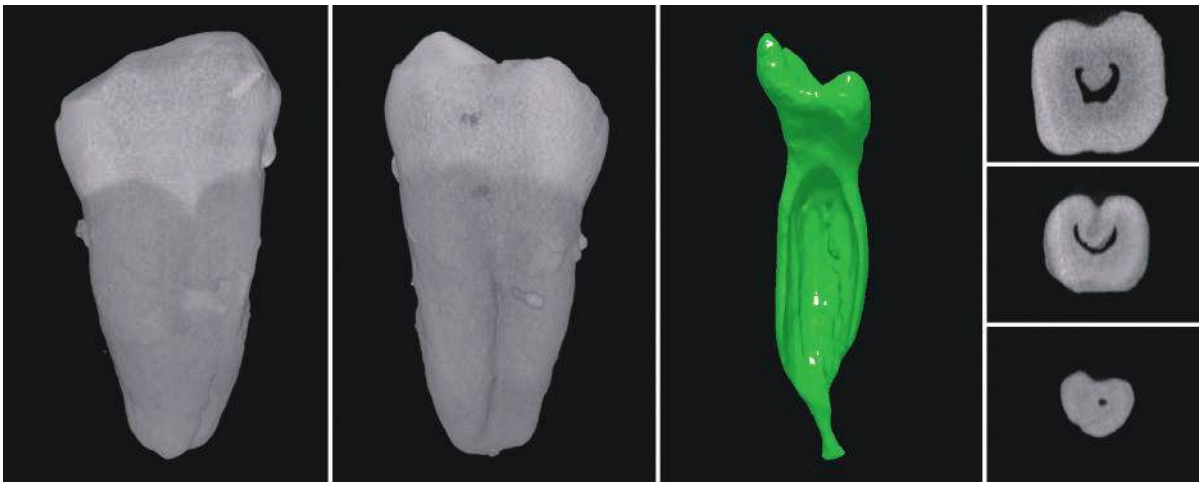


Fig. 7.11 Common variant of a C-shaped canal anatomy found in Native Americans and Asian populations.

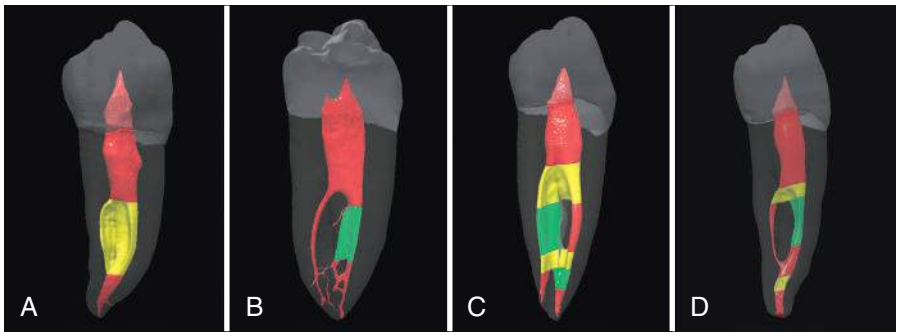


Fig. 7.12 μCT scans of mandibular premolars reflect a wide variation in canal morphology, moving from coronal to apical sections. The red color indicates a non–C-shaped canal space; green indicates a semilunar canal shape; and yellow indicates a continuous C-shaped configuration.

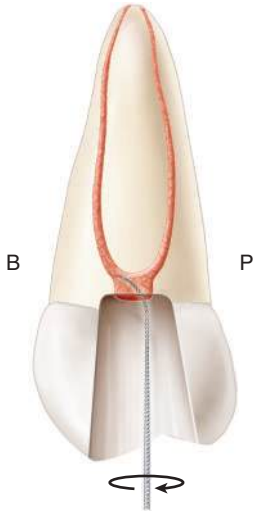


Fig. 7.13 An oval orifice must be explored with apically curved small instruments. When trying to locate the buccal canal, the clinician should place the file tip in the orifice with the tip curved to the buccal side. To explore for the palatal canal, a curved file tip is placed toward the palate. *B,* Buccal; *P,* palatal.

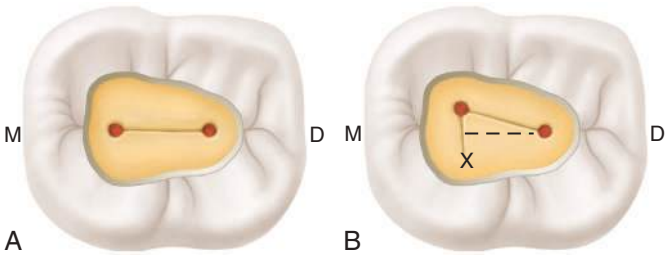


Fig. 7.14 A, In a mandibular second molar with two canals, both orifices are in the mesiodistal midline. **B,** If two orifices are not directly in the mesiodistal midline, a search should be made for another canal on the opposite side in the area of "X." *D,* Distal; *M,* mesial.

percent of the mesial roots of mandibular first molars have these communications at the apical to middle third junction, with the distal root display more in the apical third.

Another change that often occurs in the midroot portion is the splitting of a single canal into two or more canals, along with a wide variation in canal morphology. Likewise, canals often join in this area, starting coronally as two separate canals (see Fig. 7.12).

Whenever a root contains two canals that join to form one, the lingual/palatal canal generally is the one with direct access to the apex. When one canal separates into two, the division is buccal and palatal/lingual, and the lingual canal generally splits from the main canal at a sharp angle, sometimes nearly a right angle (Fig. 7.16). Although the splitting of one canal into two presents challenges, of greater concern is the very small canal that may split off, at times at almost a 90 degree angle, and cannot be located tactilely or even under magnification with the DOM. Although these canals may join the main canal at some point

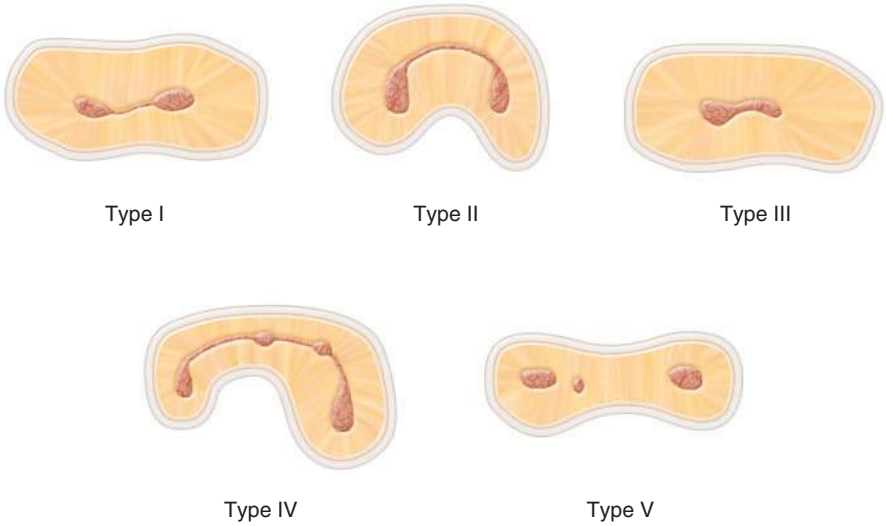


Fig. 7.15 Schematic representation of isthmus classifications described by Kim and colleagues. Type I is an incomplete isthmus; it is a faint communication between two canals. Type II is characterized by two canals with a definite connection between them (complete isthmus). Type III is a very short, complete isthmus between two canals. Type IV is a complete or incomplete isthmus between three or more canals. Type V is marked by two or three canal openings without visible connections. (From Kim S, Pecora G, Rubinstein R, Dorscher-Kim J: *Color atlas of microsurgery in endodontics*, Philadelphia, 2001, Saunders.)

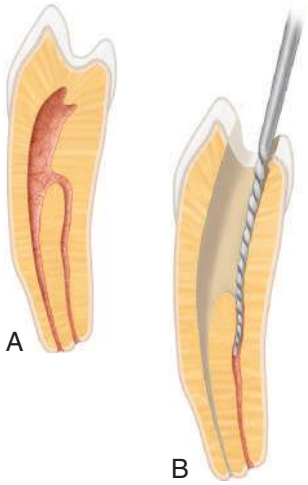


Fig. 7.16 A, Mesial view of a mandibular premolar with a type V canal configuration. The lingual canal separates from the main canal at nearly a right angle. **B,** This anatomy requires widening of access in a lingual direction to achieve straight-line access to the lingual canal. This should be done with a DOM.

apically, some of them exit separately. This is not uncommon in mandibular first premolars.

Apical Considerations

The classic concept of apical root anatomy is based on three anatomic and histologic landmarks in the apical region of a root: the apical constriction (AC), the cementodentinal junction (CDJ), and the AF. Kuttler's description of the anatomy of the root apex has the root canal tapering from the canal orifice to the AC, which generally is 0.5 to 1.5 mm coronal to the AF (Fig. 7.17).[51] The AC generally is considered the part of the root canal with the smallest diameter; it also is the reference point clinicians most often use as the apical termination for enlarging, shaping, cleaning, disinfecting, and filling. Violation of this area with instruments or filling materials is not recommended for long-term, successful outcomes.[38]

The CDJ is the point in the canal where cementum meets dentin; it is also the point where pulp tissue ends and periodontal tissues begin (Fig. 7.18). The location of the CDJ in the root canal varies considerably. It generally is not in the same area as the AC, and estimates place it approximately 1 mm from the AF.[93,101]

The AF is the "circumference or rounded edge, like a funnel or crater that differentiates the termination of the cemental canal from the exterior surface of the root."[51] The AF does not normally exit at the anatomic apex, but rather is offset 0.5 to 3 mm. This variation is more marked in older teeth through cementum apposition. Studies have shown that the AF coincides with the apical root vertex in 17% to 46% of cases.[14,35,36,87,113,115]

The location and diameter of the CDJ differ from those of the AF in maxillary anterior teeth.[88] Extension of cementum from the AF into the root canal differs considerably, even when opposite canal walls are compared (see Figs. 7.17 and 7.18). Cementum reaches the same level on all canal walls in only 5% of cases. The greatest extension generally occurs on the concave side of the canal curvature. This variability confirms that the CDJ and the AC generally are not in the same area and that the CDJ should be considered just a variable junction at which two histologic tissues meet in the root canal (see Fig. 7.18).

Scanning electron microscopy has been used to determine the number and size of main apical foramina, their distance from the anatomic apex, and the size of accessory foramina. The morphology of the apical third of the root reflects multiple anatomic variations, including numerous accessory canals; areas of resorption and repaired resorption; attached,

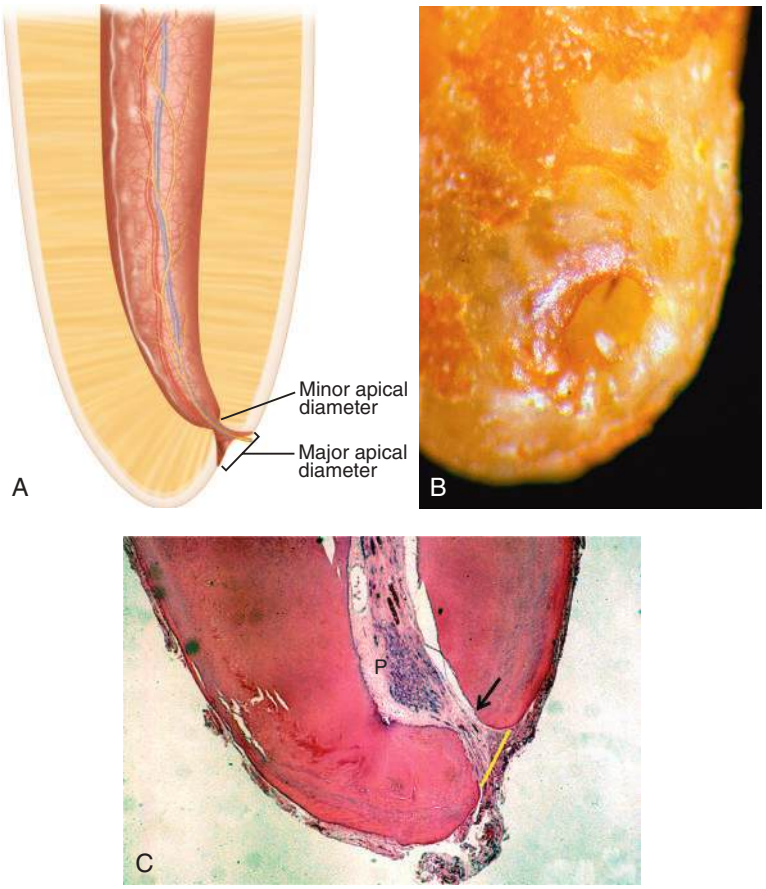


Fig. 7.17 **A**, Morphology of the root apex. From its orifice the canal tapers to the apical constriction, or minor apical diameter, which generally is considered the narrowest part of the canal. From this point the canal widens as it exits the root at the apical foramen, or major apical diameter. The space between the minor and major apical diameters is funnel shaped. **B,** Clinical view of major apical foramen. **C,** Histologic view of the apical constriction *(arrow)* and apical foramen *(yellow line)*. *P,* Pulp. (**B** and **C,** From Gutmann JL, Lovdahl PE: *Problem solving in endodontics*, ed 5, St. Louis, 2011, Elsevier.)



Fig. 7.18 Root apex following root canal filling *(RCF)* short of the actual root length. Histologic evidence shows that hard tissue *(black arrows)* has formed from the cells of the periodontal ligament *(PDL)* adjacent to the root filling material. Cementum formation on the internal aspect of the apical foramen is indicated by the yellow arrows. These findings accentuate the variable nature of the apical tissues. (From Gutmann JL, Lovdahl PE: *Problem solving in endodontics*, ed 5, St. Louis, 2011, Elsevier.)

embedded, and free pulp stones; varying amounts of reparative dentin; and varying root canal diameters.[71,72] Primary dentinal tubules are found less often than in the coronal dentin and are more or less irregular in direction and density, or even absent. This variable nature of the apical structure and significant absence of dentinal tubules may lead to reduced chances of bacterial invasion into the dentinal walls; however, it also presents challenges for all root canal procedures, from cleaning and disinfection to obturation.

Clinically, considerable controversy exists over the exact termination point for root canal procedures in the apical third of the root; clinical determination of apical canal morphology is difficult at best.[39,98] Detailed perspectives, both historical and contemporary, can be found in the literature.[38] Some authors recommend that root canal procedures terminate at or within 3 mm of the radiographic apex, depending on the pulpal diagnosis. For vital cases, clinical and biologic evidence indicates that a favorable point at which to terminate therapy is 2 to 3 mm short of the radiographic apex.[46,99] This leaves an apical pulp stump, which prevents extrusion of irritating filling materials into the periradicular tissues in vital cases. However, what is not commonly recognized among clinicians is that this so-called pulp stump is actually not pulp tissue, but rather periodontal tissue that can ensure apical healing with cementum over the root end (see Fig. 7.18). Studies have shown that a better success rate is achieved when the procedure ends at or within 2 mm of the radiographic

apex.[45,46,99] When short of the 2-mm point or extended past the radiographic apex, the success rate declined by 20%. For revision of procedural failures, apical procedures should extend to or preferably 1 to 2 mm short of the radiographic apex to prevent overextension of instruments and filling materials into the periradicular tissues.

Many investigators who have evaluated apical and periradicular tissues after root canal procedures concluded that the most favorable prognosis was obtained when procedures were terminated at the AC, and the worst prognosis was produced by treatment that extended beyond the AC.[54–57,90] Some studies support the termination of all procedures at or beyond the radiographic apex, thereby filling all apical ramifications and lateral canals.[95] This dictate is empirically based, and CBCT evaluation of procedures previously thought to be successful has allowed identification of more posttreatment disease.[18,83]

Objectives and Guidelines for Traditional Access Cavity Preparation

OBJECTIVES

Access to the complex root canal system is the first important phase of any nonsurgical root canal procedure.[39,109] The objectives of access cavity preparation are to (1) remove all caries when present, (2) conserve sound tooth structure, (3) unroof the pulp chamber completely, (4) remove all coronal pulp tissue (vital or necrotic), (5) locate all root canal orifices, and (6) achieve straight- or direct-line access to the AF or to the initial curvature of the canal.

A properly prepared access cavity creates a smooth, straight-line path to the canal system and ultimately to the apex or position of the first curvature (Fig. 7.19, *A*). Straight-line access provides the best chance of débridement of the entire canal space; it reduces the risk of instrument breakage[73]; and it results in straight entry into the canal orifice, with the line angles forming a funnel that drops smoothly into the canal (or canals). Projection of the canal center line to the occlusal surface of the tooth indicates the location of the line angles (see Fig. 7.19, *B*).

Key Steps to Consider in Traditional Access Preparation

EVALUATION OF THE CEMENTOENAMEL JUNCTION AND OCCLUSAL TOOTH ANATOMY

Traditionally, some access cavities have been prepared in relation to the occlusal or lingual anatomy. However, complete reliance on the anatomy is dangerous because this morphology can change as the crown is destroyed by caries and reconstructed with various restorative materials. Likewise, the root may not be perpendicular to the occlusal surface of the tooth; thus, complete dependence on the occlusal or lingual anatomy may explain the occurrence of some procedural errors, such as coronal perforations along the cervical line or into the furcation. Krasner and Rankow found that the cementoenamel junction (CEJ) was the most important anatomic landmark for determining the location of pulp chambers and root canal orifices. Their study demonstrated the existence of a specific and consistent anatomy of the pulp chamber floor.[49]

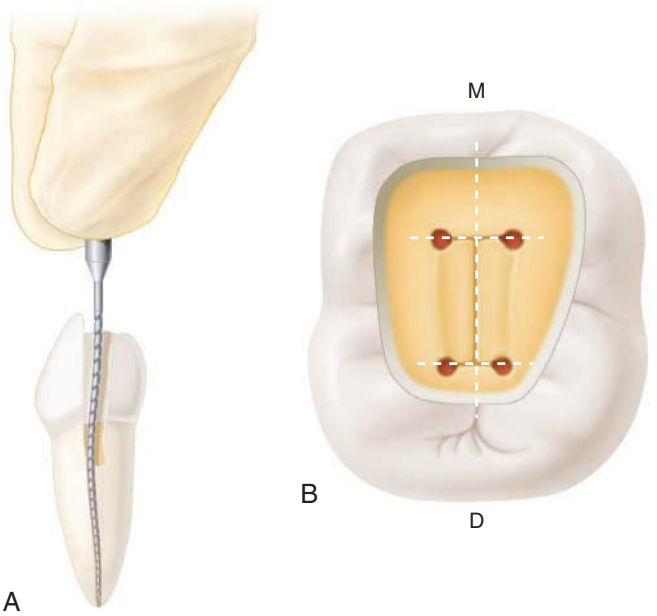


Fig. 7.19 A, Straight-line access to a canal. The instrument should not be deflected until it reaches the initial canal curvature. In some cases coronal tooth structure must be sacrificed to obtain direct access to the pulp chamber. **B,** Diagrammatic representation of centrality and concentricity of symmetry and location of canal orifices. *D,* Distal; *M,* mesial.

PREPARATION OF THE ACCESS CAVITY THROUGH THE LINGUAL AND OCCLUSAL SURFACES

Access cavities on anterior teeth usually are prepared through the lingual tooth surface, and those on posterior teeth are prepared through the occlusal surface. These approaches are the best for achieving straight-line access while reducing esthetic and restorative concerns. Some authors have recommended that the traditional anterior access for mandibular incisors be moved from the lingual surface to the incisal surface in selected cases[69]; this may allow for better access to the lingual canal and improve canal débridement (Fig. 7.20). In teeth that are lingually inclined or rotated, this is often the preferred choice for access and is performed before the dental dam is placed, or careful alignment of the root is determined before a bur is used.[39]

REMOVAL OF ALL DEFECTIVE RESTORATIONS AND CARIES BEFORE ENTRY INTO THE PULP CHAMBER

Removal of all defective restorations and caries before entering the root canal system is essential for many reasons.[39] Hidden caries or fractures are often identified, and the ability to determine the restorability of the tooth is enhanced (Fig. 7.21). In this respect, it is not uncommon, before entering the pulp chamber, to stop with the access opening preparation in favor of a crown lengthening procedure or adding a bonded buildup.[39]



Fig. 7.20 An incisal access cavity on mandibular anterior teeth may allow for improved straight-line access and canal débridement.

PREPARATION OF ACCESS CAVITY WALLS THAT DO NOT RESTRICT STRAIGHT- OR DIRECT-LINE PASSAGE OF INSTRUMENTS TO THE APICAL FORAMEN OR INITIAL CANAL CURVATURE

Sufficient tooth structure must be removed to allow instruments to be placed easily into each canal orifice without interference from canal walls, particularly when a canal curves severely or leaves the chamber floor at an obtuse angle. Hence, access design depends not only on the orifice location, but also on the position and curvature of the entire canal.[106]

Objectives and Guidelines for Minimally Invasive Access Cavity Preparation

The objectives are to preserve as much tooth structure as possible, while managing to locate, debride, enlarge, shape, and obturate the root canal(s). Much has been written and some studies have been done to verify this approach to access preparation; however, definitive, evidence-based outcomes have yet to be seen within this empirically based approach. Access openings must be crafted to preserve sound tooth structure and prevent tooth fracture (referred to as the guided endodontic technique,[33,58] or microguided technique,[23] contracted cavities,[6,66,74,97] truss access openings,[8] and orifice-directed dentin conservation access[77]). The prevention of gouging, cervically, laterally or into the floor of the pulp chamber is especially important (Figs. 7.22 and 7.23). However, access that is too restricted may or may not greatly impact the remaining and important root canal procedures (Fig. 7.24).[50,77,94] While judicious orifice location and careful canal penetration are essential, efforts should be made to minimize the excess removal of cervical tooth structure.[20–22,47,97] The literature indicates that loss of tooth structure in certain teeth may weaken the tooth cervically,[124] making them susceptible to fracture,[5,41,57,60,61] although this concept has been challenged.[91,97]

Mechanical Phases of Access Cavity Preparation

MAGNIFICATION AND ILLUMINATION

For some clinicians, the access cavity is best prepared with the use of magnification and an appropriate light source.[15] In lieu of a DOM, surgical loupes with an auxiliary light source are highly recommended.

BURS

Numerous burs have been developed exclusively for access cavity preparation (Fig. 7.25). Most clinicians have their own set of preferred access burs. A significant trend in restorative dentistry is the increased use of zirconia-based crowns and onlays. Zirconia has different mechanical and thermal characteristics than metal. Carbide burs do not efficiently or safely cut zirconia. Zirconia is a brittle material and when cut can develop cracks that propagate through the framework and lead to eventual failure of the crown or onlay. Diamond bur manufacturers are aware of these

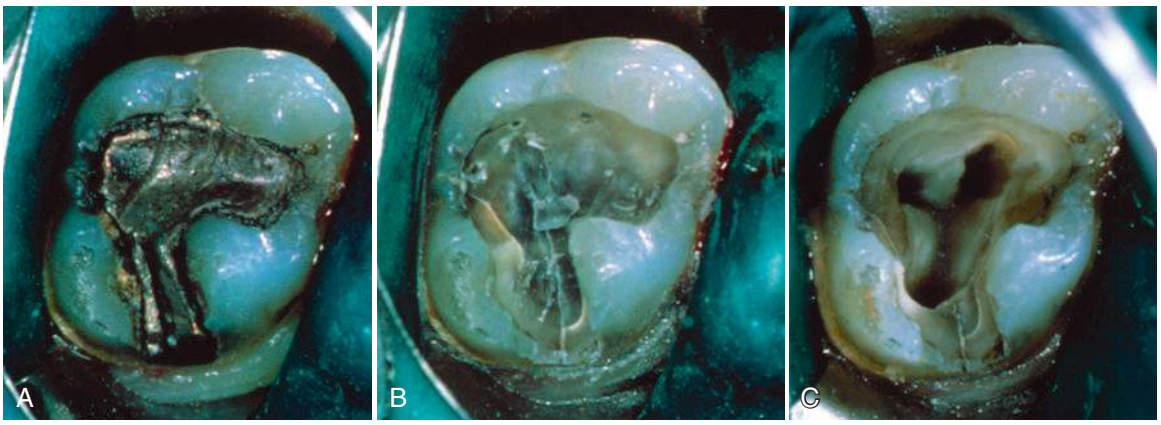


Fig. 7.21 **A,** Maxillary molar requiring root canal procedure. **B,** Removal of the amalgam reveals a vertical fracture on the palatal margin. **C,** Complete cleaning and shaping of the canals. Fracture lines are still visible, but no probings are present. (From Gutmann JL, Lovdahl PE: *Problem solving in endodontics*, ed 5, St. Louis, 2011, Elsevier.)

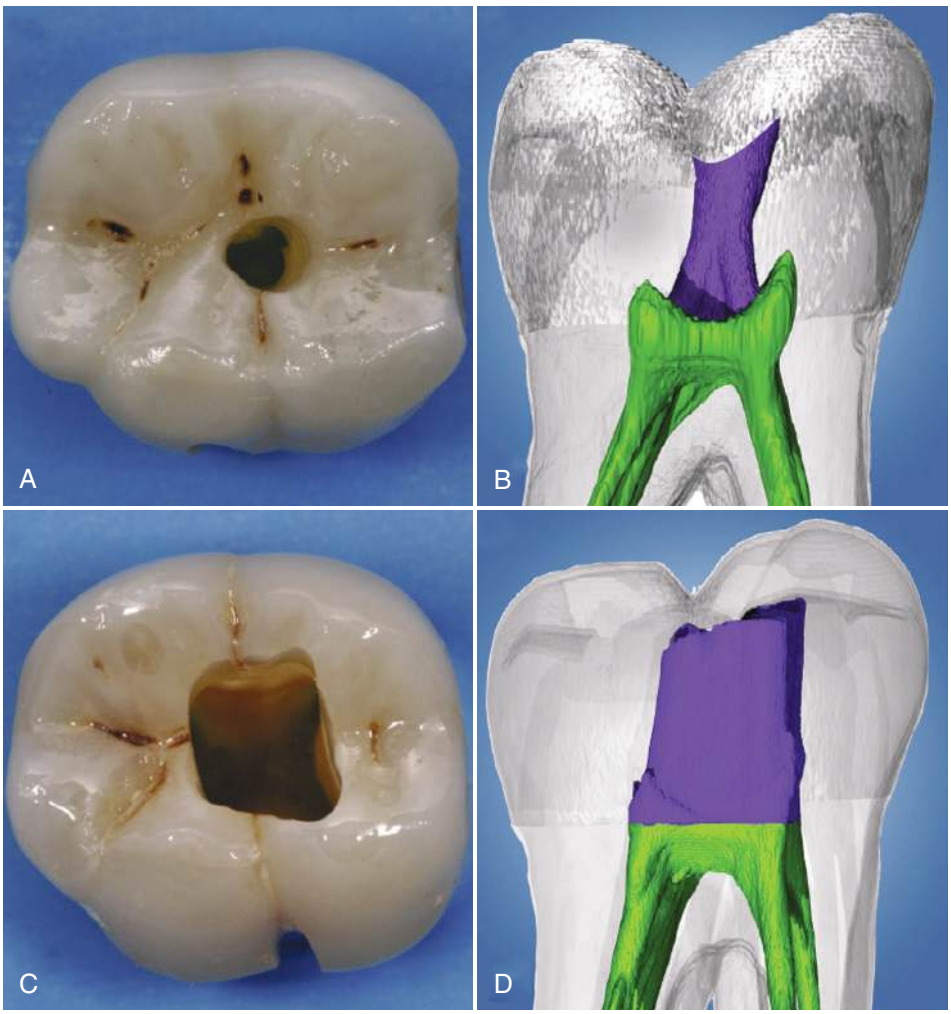


Fig. 7.22 A, A photograph of a mandibular molar showing minimally invasive access cavity outline. **B,** A sagittal view of a μCT reconstruction showing minimally invasive access cavity pathway. **C,** A photograph of a mandibular molar showing traditional access cavity outline. **D,** A sagittal view of a μCT reconstruction showing traditional access cavity pathway. (From Marchesan MA, Lloyd A, Clement DJ, et al: Impacts of contracted endodontic cavities on primary root canal curvature parameters in mandibular molars, *J Endod* 44:1558–1562, 2018.)

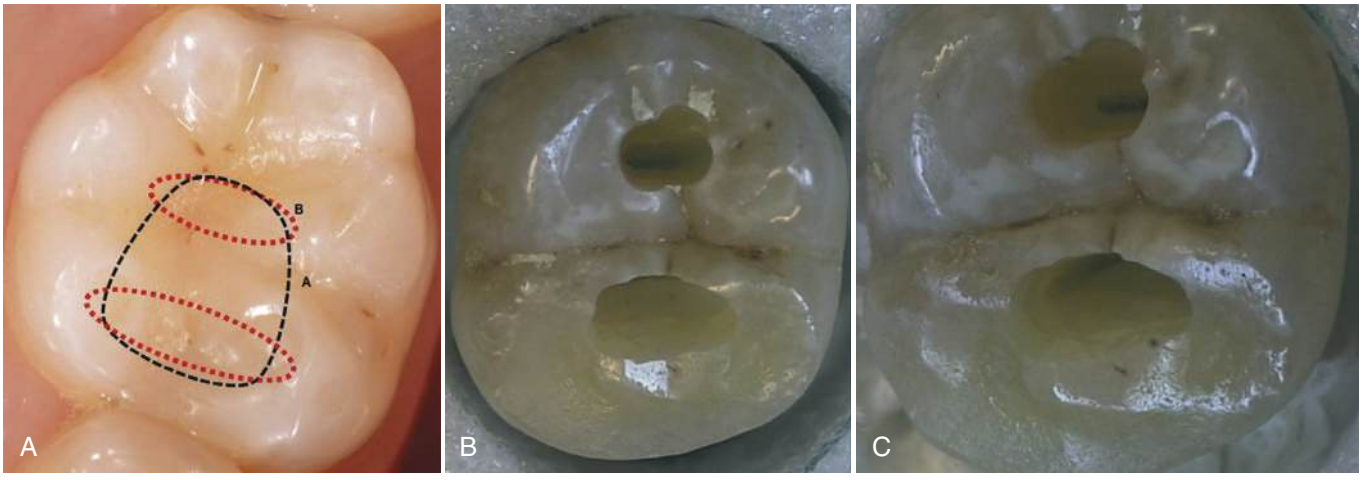


Fig. 7.23 A, A schematic diagram of the traditional access cavity outline *(A, black dotted line)* and minimally invasive access cavity outline *(B, red dotted line)* in a mandibular molar. **B,** A photograph of a mandibular molar showing mesial access and orifice. **C,** A photograph of a mandibular molar showing distal access and orifice. (From Neelakantan P, Khan K, Pak Hei G, et al: Does the orifice-directed dentin conservation access design debride pulp chamber and mesial root canal systems of mandibular molars similar to a traditional access design? *J Endod* 44:274–279, 2018.)

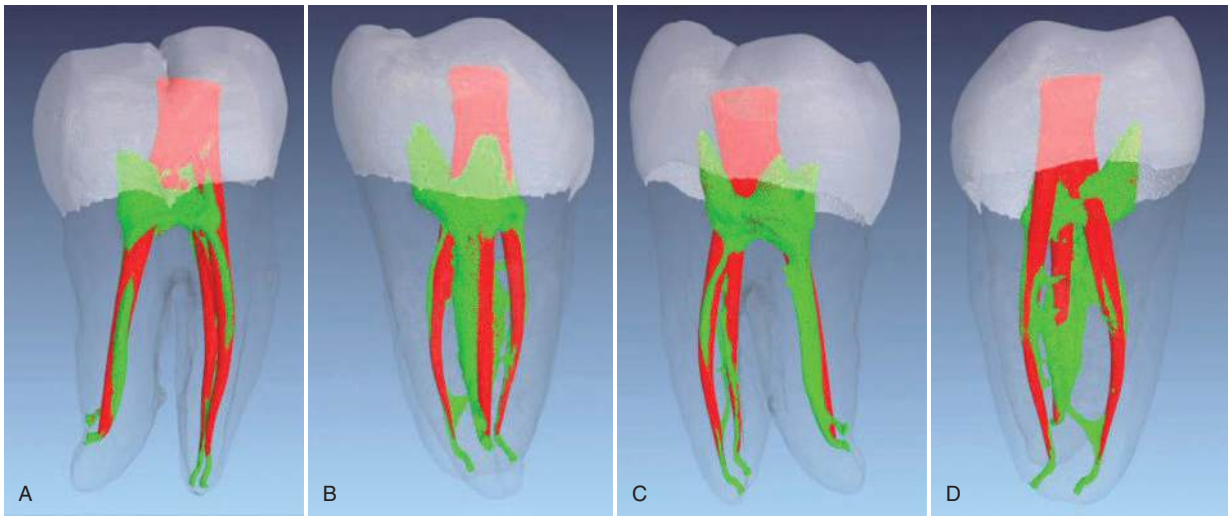


Fig. 7.24 Merged μCT images of a mandibular first molar demonstrating the root canals pretreatment *(green)* and posttreatment *(red)*. **A,** Buccal view. **B,** Distal view. **C,** Lingual view. **D,** Mesial view. (From Krishan R, Paqué F, Ossareh A, et al: Impacts of conservative endodontic cavity on root canal instrumentation efficacy and resistance to fracture assessed in incisors, premolars, and molars, *J Endod* 40:1160–1166, 2014.)



Fig. 7.25 Access burs. **A,** #2, #4, and #6 round carbide burs. **B,** Safety-tip tapered diamond bur *(left)*; safety-tip tapered carbide bur *(right)*. **C,** Transmetal bur. **D,** Round-end cutting tapered diamond bur. **E,** #57 fissure carbide bur. **F,** #2 and #4 round diamond burs. **G,** Mueller bur *(left)*; LN bur or Extendo Bur *(right)*.

issues, and they are currently introducing medium- and fine-grit diamond burs that efficiently cut zirconia (Komet USA, Savannah, GA; Prima Dental Gloucester, England; SS White). These diamond burs should be used with copious sprayed water to minimize heat buildup in the zirconia crowns during access preparations.[39]

Many teeth requiring access cavity preparations have metal restorations that must be penetrated. These restorations may be amalgams, all-metal cast restorations, or metal copings of porcelain fused to metal crowns. A transmetal bur (Dentsply Sirona, Ballaigues, Switzerland) is excellent for cutting through metal because of its exceptional cutting efficiency.

ENDODONTIC EXPLORER AND SPOON

The DG-16 endodontic explorer is used to identify canal orifices and to determine canal angulation. An alternative, the JW-17 endodontic explorer (CK Dental Industries, Orange, CA) serves the same purpose, but its thinner, stiffer tip can be useful for identifying the possible location of a calcified canal. A sharp endodontic spoon, which comes in different sizes can be used to remove coronal pulp and carious dentin.

ULTRASONIC UNIT AND TIPS

Ultrasonic units and tips specifically designed for activities during access opening preparation are extremely valuable. Ultrasonic tips from various manufacturers can be used to trough and deepen developmental grooves, remove tissue, and explore for canals. These systems provide outstanding visibility compared with traditional handpiece heads, which typically obstruct vision.

Access Cavity Preparations

ANTERIOR TEETH

The following discussion outlines the steps for maxillary and mandibular anterior teeth. Tooth-specific access concerns are illustrated and discussed in the section Morphology of and Access Cavity Preparations for Individual Teeth, later in the chapter.

External Outline Form

For an intact tooth, cutting commences at the center of the lingual or palatal surface of the anatomic crown. An outline form is developed that is similar in geometry to an ideal access shape for the particular anterior tooth. The bur is directed perpendicular to the lingual surface as the external outline opening is created.

Penetration of the Pulp Chamber Roof (Fig. 7.26)

Experienced clinicians usually penetrate the pulp chamber roof using a high-speed handpiece; however, less experienced clinicians may find the increased tactile sensation of

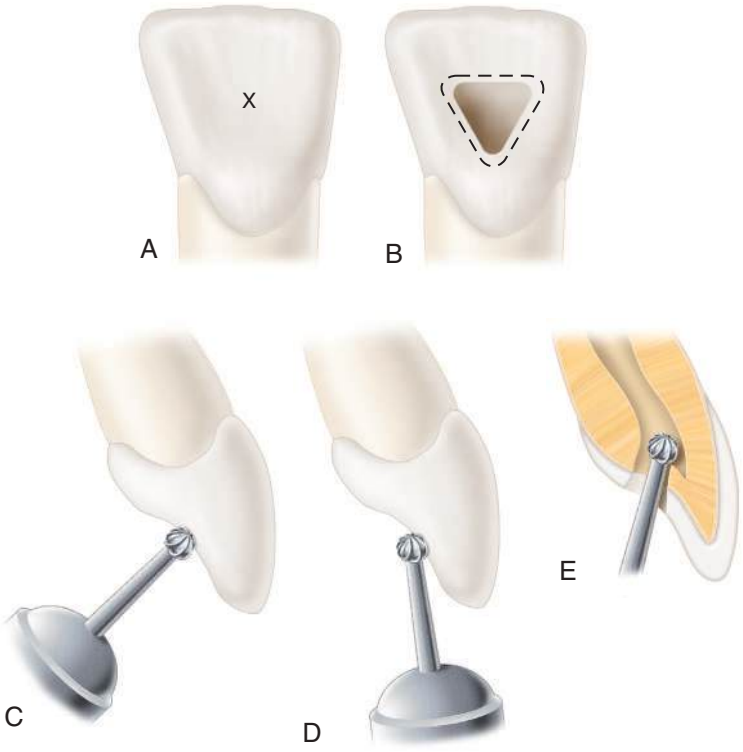


Fig. 7.26 A, In anterior teeth, the starting location for the access cavity is the center of the anatomic crown on the lingual surface *(X)*. **B,** Preliminary outline form for anterior teeth. The shape should mimic the expected final outline form, and the size should be half to three fourths the size of the final outline form. **C,** The angle of penetration for the preliminary outline form is perpendicular to the lingual surface. **D,** The angle of penetration for initial entry into the pulp chamber is nearly parallel to the long axis of the root. **E,** Completion of removal of the pulp chamber roof; a round carbide bur is used to engage the pulp horn, cutting on a lingual withdrawal stroke.

a slow-speed handpiece a safer option. Continuing with the same round or tapered fissure bur, the angle of the bur is rotated from perpendicular to the lingual/palatal surface to parallel to the long axis of the root (see Fig. 7.26, *C* and *D*). Penetration into the tooth continues along the root's long axis until the roof of the pulp chamber is penetrated; frequently, a drop into the chamber effect is felt when this occurs. Measuring the distance from the incisal edge to the roof of the pulp chamber on a dimensionally accurate pretreatment radiograph may serve as guide in limiting penetration and possibly preventing a perforation. If the drop-in effect is not felt at this depth, a DG-16 explorer can be used to probe the depth of the access. Often a small opening into the chamber is present, or the dentin is very thin and the explorer penetrates into the chamber. The depth and angle of penetration should be assessed for any deviation away from the long axis of the root in both the mesiodistal and buccolingual dimensions, and the penetration angle should be realigned if necessary. Angled radiographs can be used to assess progress at any time if any confusion or doubt exists.

Removal of the Chamber Roof

Once the pulp chamber has been penetrated, the remaining roof is removed by catching the end of a round bur under the lip of the dentin roof and cutting on the bur's withdrawal stroke (see Fig. 7.26, *E*). Working in this manner enables the internal pulp anatomy to dictate the external outline form of the access opening. Copious irrigation with sodium hypochlorite helps control any hemorrhage present. Subsequently, the chamber roof, including the pulp horns, is removed and all internal walls are flared to the lingual surface of the tooth and complete roof removal is confirmed with an explorer.

Removal of the Lingual Shoulder and Coronal Flaring of the Orifice

Once the orifice or orifices have been identified and confirmed, the lingual shoulder or ledge is removed. This is a shelf of dentin that extends from the cingulum to a point approximately 2 mm apical to the orifice (Fig. 7.27). Its removal improves straight-line access and allows for more intimate contact of files with the canal's walls for effective shaping and cleaning. Furthermore, its removal from mandibular anterior teeth may often expose an extra orifice and canal.

The contemporary approach to flaring the orifice involves the use of rotary nickel-titanium (NiTi) orifice openers that allow rapid, safe removal of the lingual ledge, following the manufacturer's directions for use (DFUs). Some clinicians prefer the use of Gates-Glidden burs. These openers allow refinement of the orifice shape or help to enhance straight-line access to the canal with minimal removal of dentin when used properly. The bur should be placed so as to avoid putting a bevel on the incisal edge of the access preparation (Fig. 7.28). When Gates-Glidden burs are used, the largest that can be placed passively 2 mm apical to the orifice is used first with gentle pressure. During this process, the orifice is often concomitantly flared so that it is contiguous with all walls of the access preparation. If this is not achieved, use of the orifice openers is recommended.

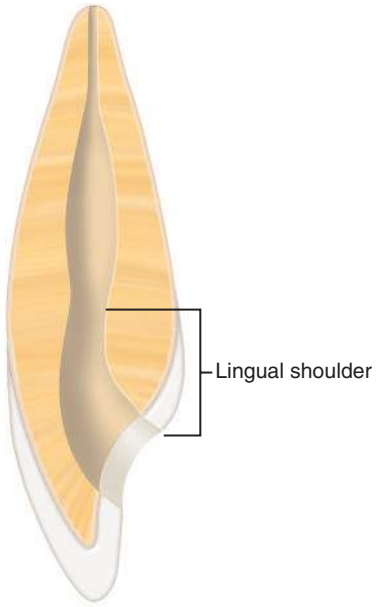


Fig. 7.27 Lingual shoulder of the anterior tooth, extending from the cingulum to 2 mm apical to the orifice.



Fig. 7.28 Placing an incisal bevel on the lingual surface of a maxillary anterior tooth can lead to fracture of the permanent restoration during occlusal function.

Straight-Line Access Determination

Straight-line access to the canal orifice is the ideal goal. Preferably, a small intracanal file should reach the AF or the first point of canal curvature with no deflections. If the lingual shoulder has been removed and a file still binds on the incisal edge, the access cavity should be further extended incisally, until the file is not deflected (Fig. 7.29). The final position of the incisal wall of the access cavity is determined by two factors: (1) complete removal of the pulp horns and (2) straight-line access. However, the concept of



Fig. 7.29 A, The lingual ledge of dentin remains, deflecting the file toward the labial wall. As a result, portions of the lingual canal wall will not be shaped and cleaned. **B,** Removal of the lingual ledge results in straight-line access.

straight-line access may not be totally feasible when using minimally invasive access preparations.

Refinement and Smoothing of Restorative Margins

The final step in the preparation of an access cavity is to refine and smooth the cavosurface margins. Rough or irregular margins can contribute to coronal leakage through a permanent or temporary restoration.

Individual Anterior Teeth

See the figures in the section Morphology of and Access Cavity Preparations for Individual Teeth, later in this chapter.

POSTERIOR TEETH

Preparing access cavities on posterior teeth is similar to the process for anterior teeth, but significant differences warrant a separate discussion.[39] Posterior teeth requiring root canal procedures typically have been heavily restored, which may make the process more challenging, or the carious process was extensive, which may make the process much easier.

External Outline Form

Removal of caries and existing restorations from a posterior tooth often results in the development of an acceptable access outline form, with pulp chamber positioned in the center of the tooth at the level of the CEJ. Therefore, in maxillary premolars, the point of entry that determines the external outline form is on the central groove between the cusp tips (Fig. 7.30). Crowns of mandibular premolars are tilted lingually relative to their roots (Fig. 7.31); therefore, the starting location must be adjusted to compensate for this tilt (Fig. 7.32). In mandibular first premolars, the starting location is halfway up the lingual incline of the buccal cusp on a line connecting the cusp tips. Mandibular second premolars require less of an adjustment because they have less lingual inclination. The starting location for this tooth

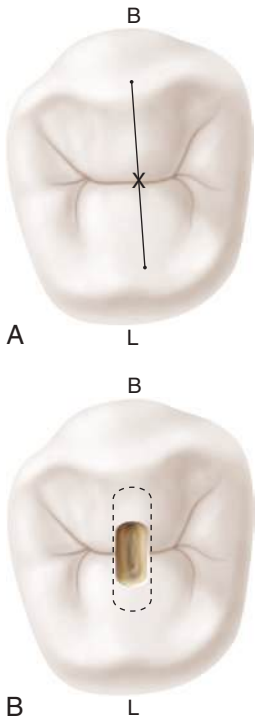


Fig. 7.30 A, Starting location for access to the maxillary premolar *(X)*. **B,** Initial outline form *(dark area)* and projected final outline form *(dashed line)*. *B,* Buccal; *L,* lingual.

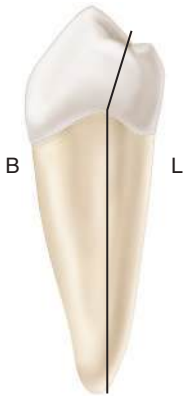


Fig. 7.31 The crown of a mandibular premolar is tilted lingually relative to the root. *B,* Buccal; *L,* lingual.

is one third the way up the lingual incline of the buccal cusp on a line connecting the buccal cusp tip and the lingual groove between the lingual cusps.

To determine the starting location for molar access cavity preparations, the mesial-distal and apical-coronal boundary limitations for this outline must be determined (Fig. 7.33). Evaluation of bite-wing radiographs is an accurate method of assessing the extensions of the pulp chamber (Fig. 7.34). The mesial boundary for both the maxillary and mandibular molars is a line connecting the mesial cusp tips. Pulp chambers are rarely found mesial to this imaginary line. A good initial distal boundary for maxillary molars is the oblique ridge. For mandibular molars, the initial distal boundary is a line connecting the buccal and lingual grooves. For molars, the correct

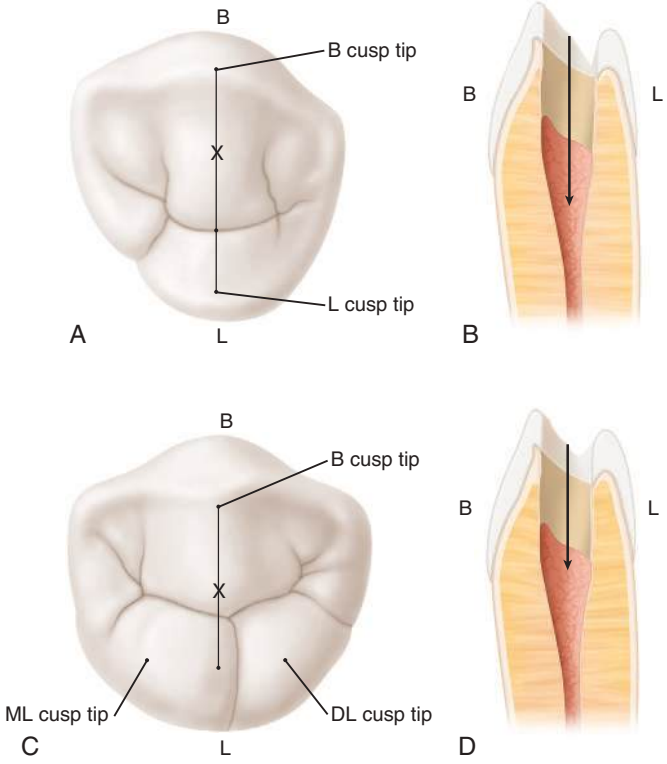


Fig. 7.32 A, Mandibular first premolar and access starting location *(X)* (occlusal view). **B,** Mandibular first premolar and starting location (proximal view). **C,** Mandibular second premolar and access starting location *(X)* (occlusal view). **D,** Mandibular second premolar and starting location (proximal view). *B,* Buccal; *DL,* distolingual; *L,* lingual; *ML,* mesiolingual.

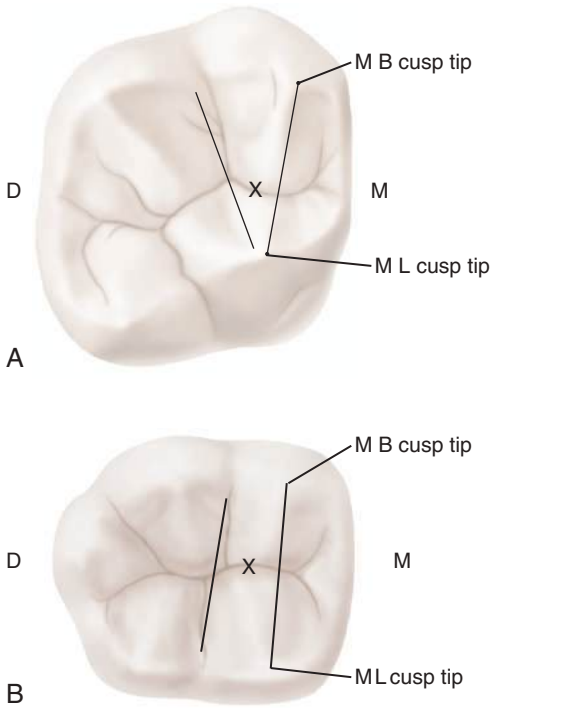


Fig. 7.33 A, Mesial and distal boundary of a maxillary molar with the access starting location *(X)*. **B,** Mesial and distal boundary of a mandibular molar showing the access starting location *(X)*. *D,* Distal; *M,* mesial; *MB,* mesiobuccal; *ML,* mesiolingual.

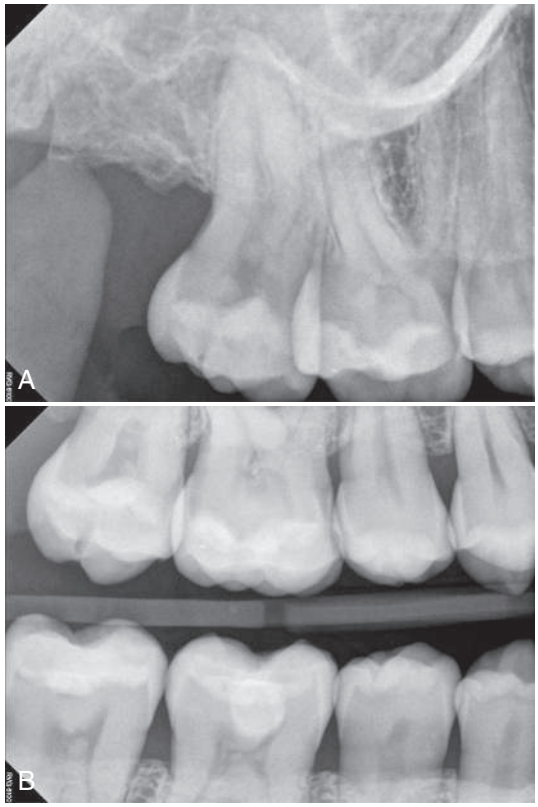


Fig. 7.34 A, Periapical radiograph of the teeth in the maxillary right quadrant. **B,** Bitewing radiograph of the same teeth provides a clearer delineation of the pulpal morphology.

starting location is on the central groove halfway between the mesial and distal boundaries.

Penetration through the enamel into the dentin (approximately 1 mm) is achieved with a #2 or #4 round bur for premolars and a #4 or #6 round bur for molars. A tapered fissure bur may be used instead of round burs. The bur is directed perpendicular to the occlusal table, and an initial outline shape is created at about half to three fourths its projected final size. The premolar shape is oval and widest in the buccolingual dimension. The molar shape is also oval initially; it is widest in a buccolingual dimension for maxillary molars and in a mesiodistal direction for mandibular molars. The final outline shape for molars is approximately triangular (for three canals) or rhomboid (for four canals), even when a minimal access approach is used; however, the canal orifices dictate the position of the corners of these geometric shapes. Therefore, until the orifices have been located, the initial outline form should be left as roughly oval.

Penetration of the Pulp Chamber Roof

As with anterior teeth, penetration through the chamber roof is limited to the distance measured on a pretreatment radiograph. If the drop-in effect is not felt at this depth, a careful evaluation of the angle of penetration is necessary before going deeper into the chamber. In multirooted posterior teeth, lateral and furcation perforations may occur rapidly without attention to 3D detail during this penetration. Aggressive probing with a DG-16 explorer at any time during the penetration is encouraged. Once the pulp

chamber has been penetrated, the remaining roof is removed by catching the end of a round bur under the lip of the dentin roof and cutting with the burs on a withdrawal stroke.

Identification of All Canal Orifices

The canal orifices play an important role in determining the final extensions of the external access outline form. Ideally, the orifices are located at the corners of the final preparation to facilitate all of the root canal procedures (Fig. 7.35).

Removal of the Cervical Dentin Bulges and Orifice and Coronal Flaring

Internal impediments to an ideal access opening are the cervical dentin ledges or bulges and the natural coronal canal constriction in posterior teeth.[39] The cervical bulges are shelves of dentin that frequently overhang orifices in posterior teeth, restricting access into root canals and accentuating existing canal curvatures (Fig. 7.36).[60]

Straight-Line Access Determination

Straight-line access is paramount to successful shaping, especially given the complexity of the root canal systems in posterior teeth. However, this has not been verified when using a minimal access cavity. Files must have unimpeded access to the AF or the first point of canal curvature to perform properly.

Visual Inspection of the Pulp Chamber Floor

Floor and walls must be inspected, using appropriate magnification and illumination, to ensure that all canal orifices are visible and no roof overhangs are present (Fig. 7.37).

Individual Posterior Teeth. See the figures in the section Morphology of and Access Cavity Preparations for Individual Teeth, later in the chapter.



Fig. 7.35 An endodontic explorer is used to search for canal orifices.

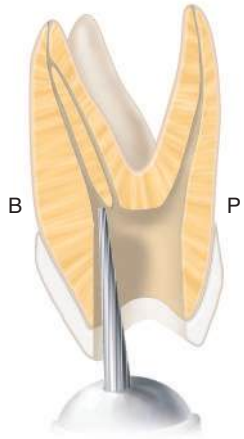


Fig. 7.36 Safety-tip carbide bur is used to shape the axial wall in one plane from the orifice to the cavosurface margin. *B,* Buccal; *P,* palatal.

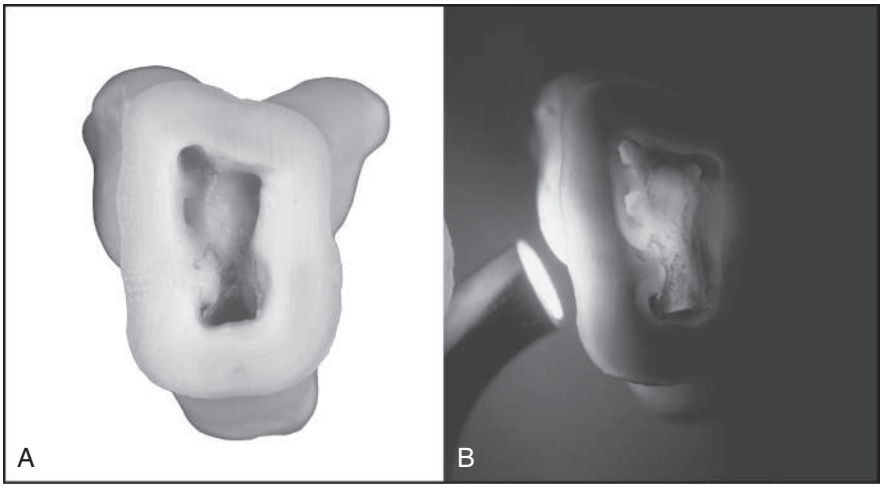


Fig. 7.37 A fiberoptic light can be applied to the cervical aspect of the crown to help obtain maximal visibility with magnification. Transillumination often reveals landmarks otherwise invisible to the unaided eye.

Challenging Access Preparations

ACCESS IN TEETH WITH PRESUMED CALCIFIED CANALS

Chambers and roots that demonstrate the potential for significant calcifications may present problems with locating, penetrating, and negotiating the pulpal spaces.[39] The use of magnification and transillumination, as well as careful examination of color changes and pulp chamber shapes, assists in locating the canals (Fig. 7.38). However, the search for the root canal orifices should be made only after the pulp chamber has been completely prepared and its floor has been cleaned and dried (95% denatured ethanol may be useful for drying the floor and enhancing visibility). A fiberoptic light directed from the external through the CEJ can reveal subtle landmarks and color changes that may not otherwise be visible. The chamber floor is darker in color than its walls, and developmental grooves connecting orifices are lighter in color than the chamber floor. Awareness of these color differences when searching for small orifices is essential, especially when searching for those that are located at the angles formed by the floor and walls and at the end points of developmental grooves. Additional methods to help locate calcified root canals include staining the pulp chamber floor with 1% methylene blue dye, performing the sodium hypochlorite "champagne bubble" test (Fig. 7.39), and searching for canal bleeding points. These approaches are enhanced when the area is viewed through magnification, which would be essential with a minimal access cavity.

Uncovering canals that contain calcified material is a challenge. When the canal is located, a small K-file (#6, #8, or #10 or, preferably, a C or C+ file [Dentsply Sirona—Ballaigues, Switzerland]) coated with a chelating agent should be introduced into the canal to determine patency. These instruments provide added stiffness to the shaft for better penetration. The file should not be removed until some canal enlargement has occurred. The file should be used in short up-and-down movements and in a selective circumferential filing motion, with most of the lateral pressure directed away from the furcation. This enlarges the coronal aspect of the canal safely and moves it laterally, to avoid thinning of the dentin wall adjacent to the furcation. It also creates a path of insertion for larger files and for coronal access burs. Figs. 7.40–7.45 illustrate several methods that can be used to locate calcified spaces. Furthermore, a guided access technique is advocated to perform the coronal entry into teeth with partially or completely calcified pulp chambers and/or obstructed root canals.[59]



Fig. 7.39 Allowing sodium hypochlorite (NaOCl) to remain in the pulp chamber may help locate a calcified root canal orifice. Tiny bubbles may appear in the solution, indicating the position of the orifice. This is best observed through magnification.

CROWDED OR ROTATED TEETH

Traditional or minimal access preparations may not be possible in patients with crowded or rotated teeth. The decision about an alternative approach must be based on straight-line access principles and conservation of tooth structure.[58] In certain circumstances, a buccal or incisal access preparation may be indicated, or developing the standard shape based on the degree of rotation (Fig. 7.46).[39]



Fig. 7.38 Mandibular molar with what appears to be almost complete calcification of the pulp chamber and root canals. However, pathosis is present, which indicates the presence of bacteria and some necrotic tissue in the apical portion of the roots.

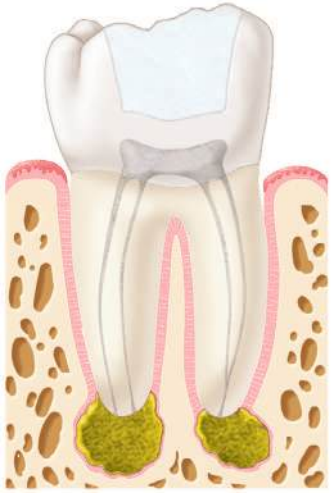


Fig. 7.40 Mandibular first molar with a class I restoration, calcified canals, and periradicular radiolucencies. Presumably a pulp exposure has occurred, resulting in calcification and ultimate necrosis of the pulp tissue.

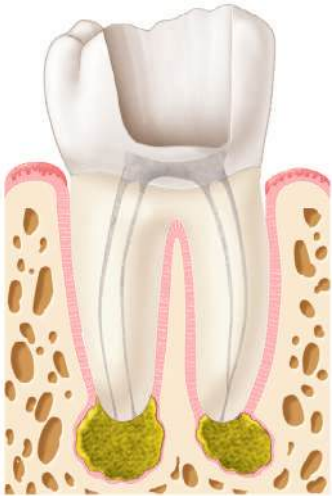


Fig. 7.41 Excavation of a restoration and base material. The cavity preparation is extended toward the assumed location of the pulp chamber, keeping in mind that pulp chambers are located in the center of the tooth at the level of the cementoenamel junction (CEJ).

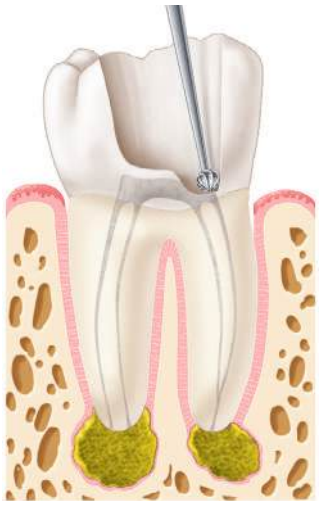


Fig. 7.42 Use a long-shank #2 or #4 round bur to remove dentin and attempt to locate calcified canals.

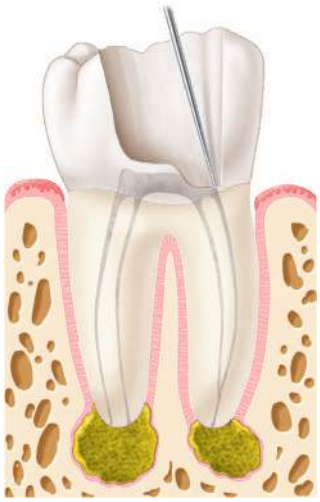


Fig. 7.43 An endodontic explorer is used to probe the pulp floor. A straight ultrasonic tip may be used to remove dentin. Angled radiographs must be taken to monitor progress.

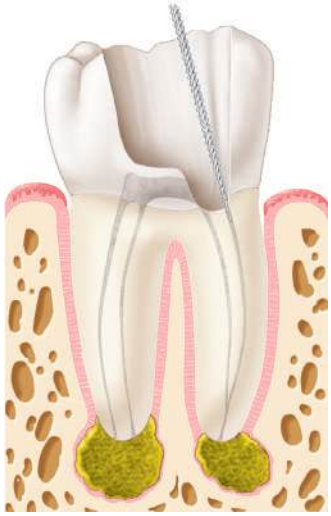


Fig. 7.44 At the first indication of a canal space, the smallest instrument (i.e., a #.06 or #.08 C or C+ file or micro openers) should be introduced into the canal. Gentle passive movement, both apical and rotational, often produces some penetration. A slight pull, signaling resistance, usually is an indication that the canal has been located. This should be confirmed by radiographs.

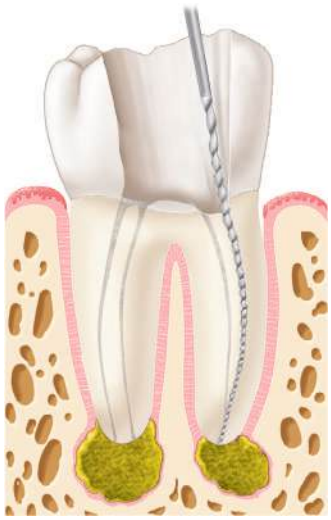


Fig. 7.45 A small K-file negotiates the canal to its terminus. An apex locator or radiograph is used to confirm the file's position.

Other problems can occur when tooth angulations are not considered during access development, such as:

- Mistaken identification of an already located canal, resulting in a search in the wrong direction for additional canals. Whenever a difficult canal is located, a file should be placed in the canal and an angled radiograph taken.
- Failure to locate a canal or extra canals.
- Excessive gouging of coronal or radicular tooth structure.
- Instrument separation during attempts to locate an orifice.
- Failure to débride all pulp tissue from the chamber.



Fig. 7.46 A, Access cavity on crowded mandibular anterior teeth. The access preparation is cut through the buccal surface on the canine. The lateral incisor has also been accessed through the buccal surface; root canal procedures were performed, and the access cavity was permanently restored with composite. **B,** Obturation.

Morphology of and Access Cavity Preparations for Individual Teeth

The anatomy shown in the following figures was obtained from human teeth through the use of recently developed 3D imaging techniques. The teeth were scanned in a high-resolution, microcomputer-assisted tomographic scanner. The data were then manipulated with proprietary computer programs to produce the 3D reconstructions and visualization. The following individuals and sources are recognized to have contributed greatly to this endeavor.

MAXILLARY CENTRAL INCISOR

The root canal system outline of the maxillary central incisor reflects the external surface outline (Fig. 7.47). The μCT scans of this tooth are seen in Fig. 7.48. The external access outline form for the maxillary central incisor is a rounded triangle with its base toward the incisal aspect (Fig. 7.49).

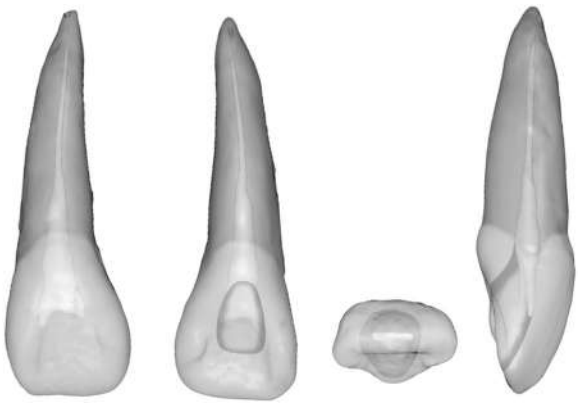


Fig. 7.47 Maxillary central incisor. Average time of eruption—7 to 8 years; average age of calcification—10 years; average length—22.5 mm. Root curvature (most common to least common): straight, labial, distal.

MAXILLARY LATERAL INCISOR

The pulp chamber outline of the maxillary lateral incisor is similar to that of the maxillary central incisor; however, it is smaller, and two or no pulp horns may be present (Fig. 7.50). The μCT scans of this tooth are shown in Fig. 7.51.

MAXILLARY CANINE

The root canal system of the maxillary canine is similar in many ways to that of the maxillary incisors (Fig. 7.52). A major difference is that it is wider labiolingually than mesiodistally. Another difference is that it has no pulp horns. Usually one root canal is present, although two canals have been reported. The μCT scans for the maxillary canine can be seen in Fig. 7.53.

MAXILLARY FIRST PREMOLAR

Most maxillary first premolars have two root canals, regardless of the number of roots (Fig. 7.54). Ethnicity plays a factor in that Asian people have a higher incidence of one canal than do other ethnic groups.[62,81] A furcation groove or developmental depression on the palatal aspect of the buccal root is another anatomic feature. Its prevalence has been reported as 62% to 100%.[42,53,108] At the deepest part of the invagination, the average dentin thickness was found to be 0.81 mm. The pulp chamber of the maxillary first premolar is considerably wider buccolingually than mesiodistally. In the buccolingual dimension, the chamber outline shows a buccal and a palatal pulp horn. The buccal pulp horn usually is larger. The maxillary first premolar may have one, two, or three roots and canals; it most often has two. The μCT scans for the maxillary first premolar can be seen in Fig. 7.55.

MAXILLARY SECOND PREMOLAR

The root canal system of the maxillary second premolar is wider buccolingually than mesiodistally (Fig. 7.56). This tooth may have one, two, or three roots and canals. Two or

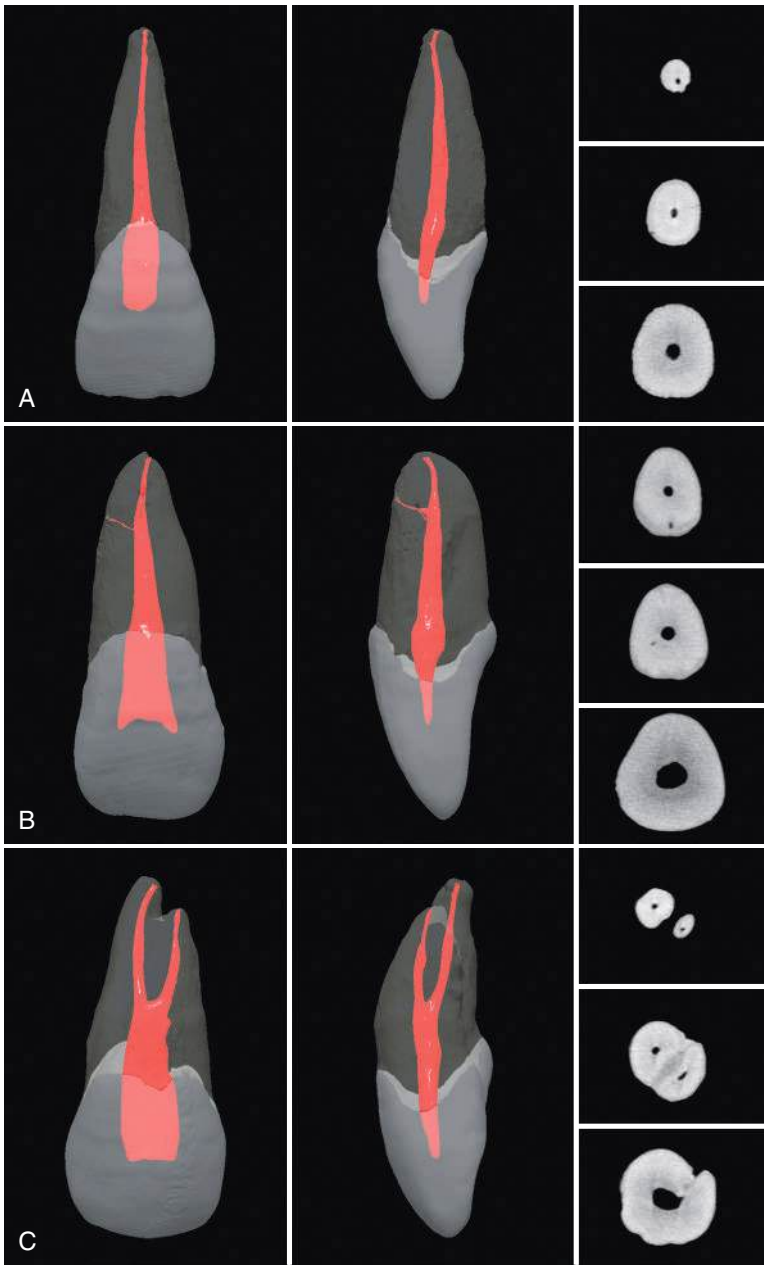


Fig. 7.48 μCT scans of maxillary central incisors. **A,** Common anatomic presentation. **B,** Central incisor with a lateral canal, which is common. **C,** Rare multiple-canal variation. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

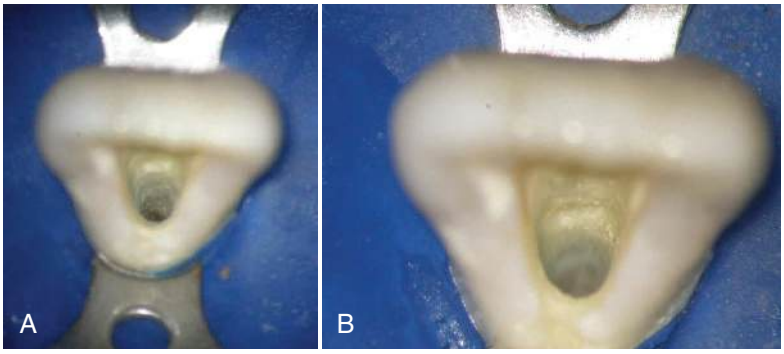


Fig. 7.49 Access cavity for a maxillary central incisor as viewed through the dental operating microscope. **A,** ×3.4 magnification. **B,** ×8.4 magnification.

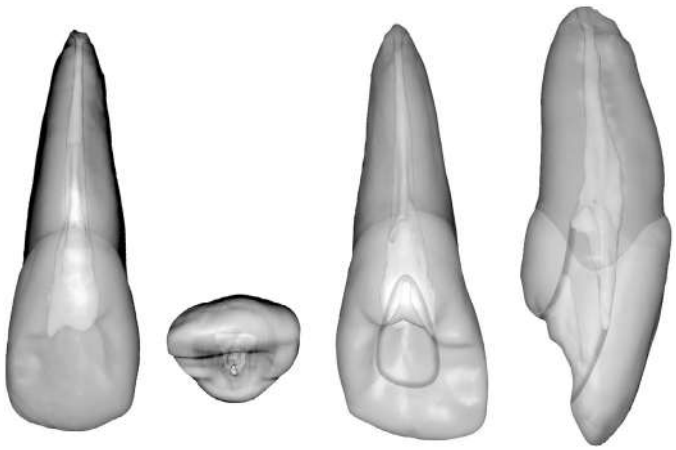


Fig. 7.50 Maxillary lateral incisor. Average time of eruption—8 to 9 years; average age of calcification—11 years; average length—22 mm. Root curvature (most common to least common): distal, straight.

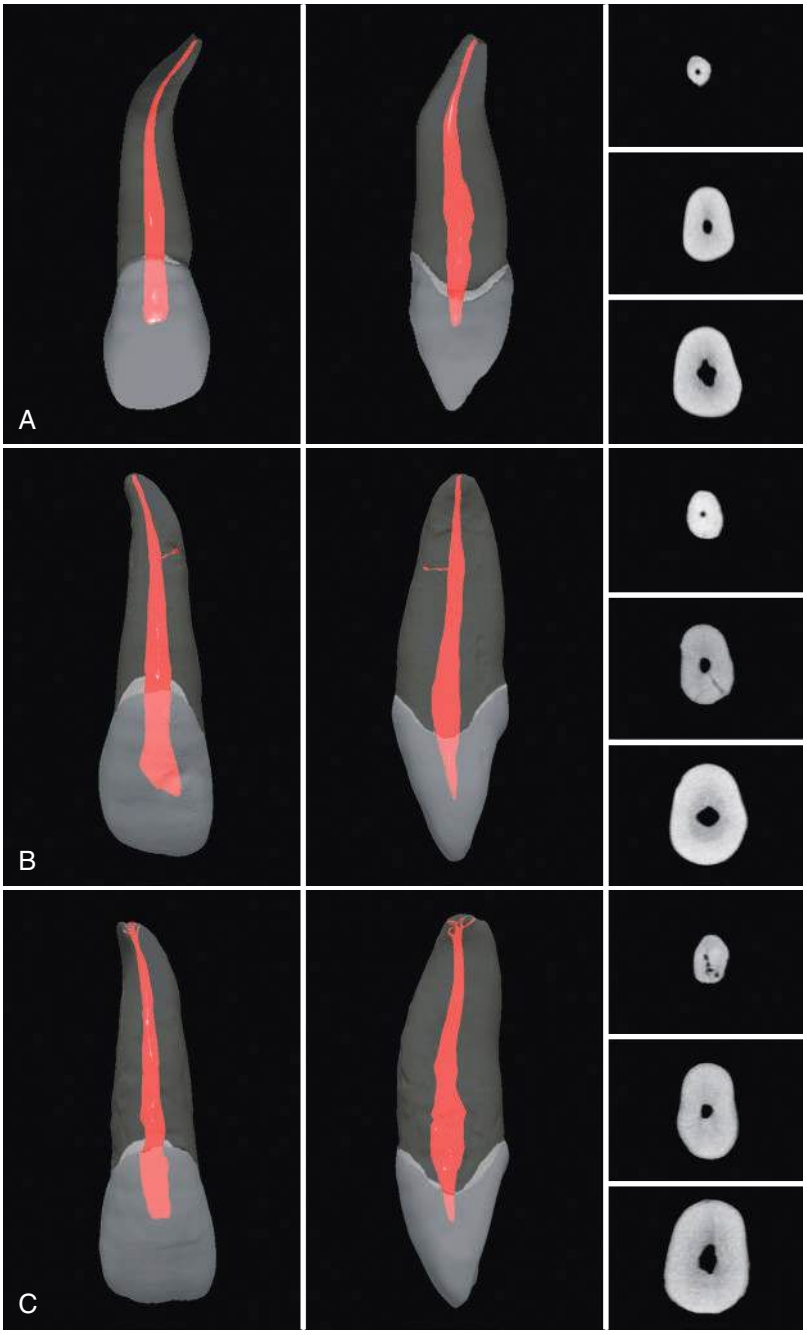


Fig. 7.51 μCT scans of maxillary lateral incisors. **A,** Common anatomic presentation. **B,** Lateral incisor with a large lateral canal, which is common. **C,** Lateral incisor with an apical delta. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

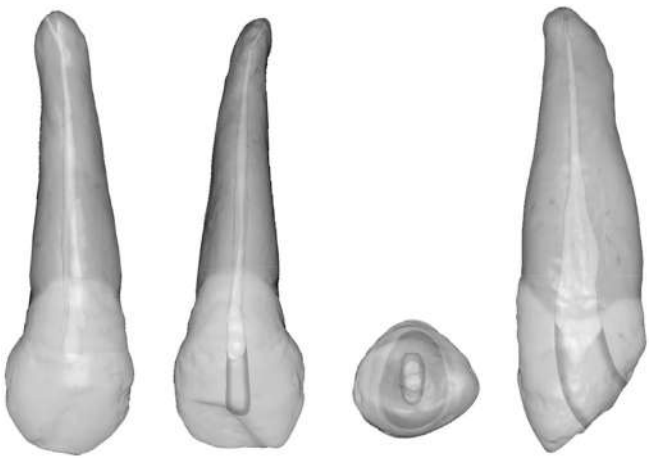


Fig. 7.52 Maxillary canine. Average time of eruption—10 to 12 years; average age of calcification—13 to 15 years; average length—26.5 mm. Root curvature (most common to least common): distal, straight, labial.

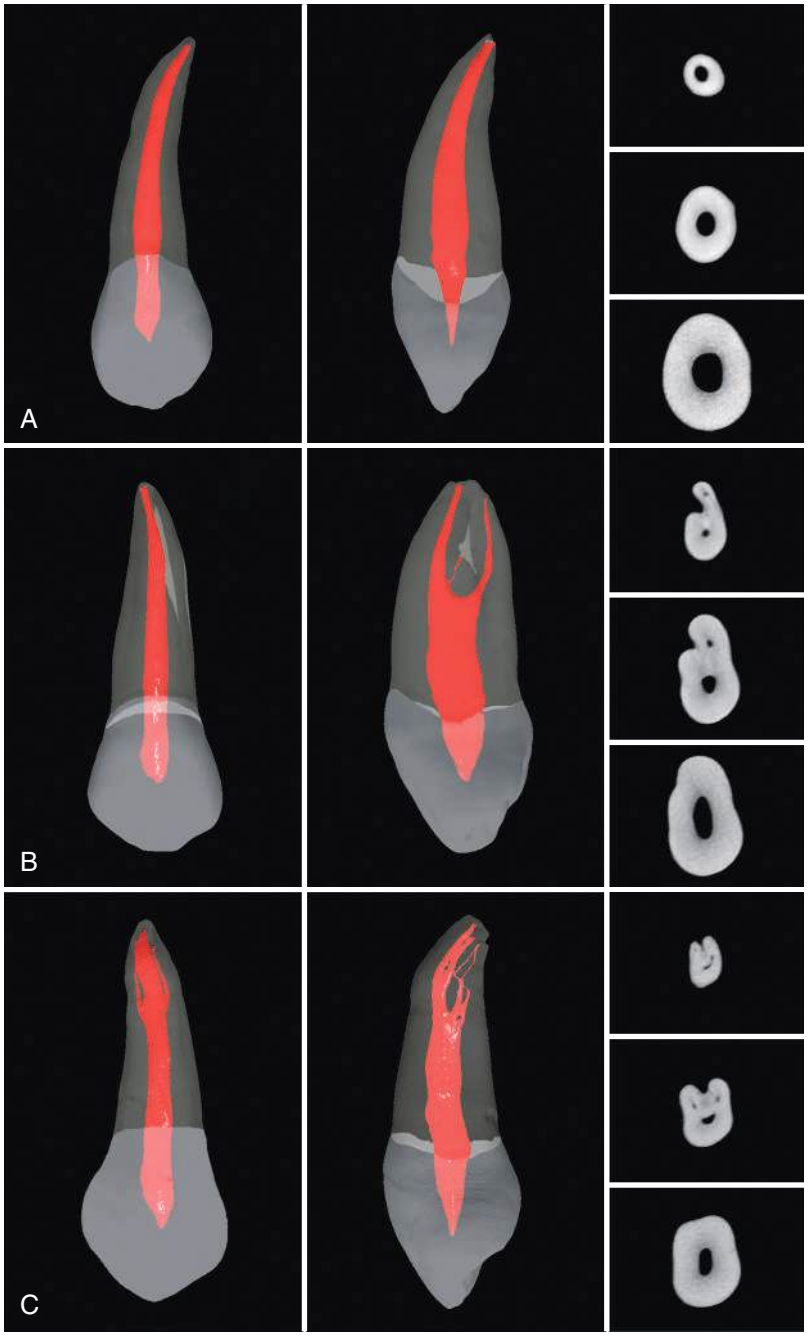


Fig. 7.53 μCT scans for the maxillary canine. **A,** Common anatomic presentation. **B,** Canine with two roots. **C,** Canine with significant deviations of the canal system in the apical third. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

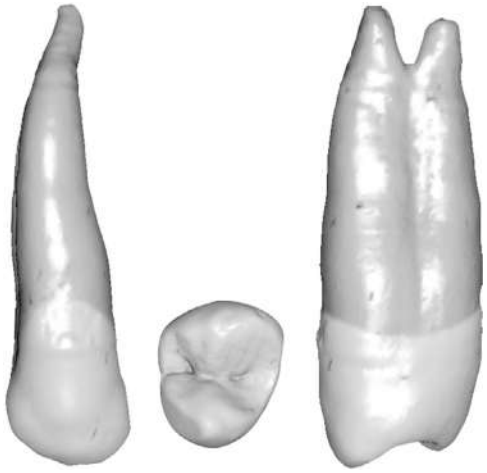


Fig. 7.54 Maxillary first premolar. Development and anatomic data: average time of eruption—10 to 11 years; average age of calcification—12 to 13 years; average length—20.6 mm. Root curvature (most common to least common): buccal root—lingual, straight, buccal; palatal root—straight, buccal, distal; single root—straight, distal, buccal.

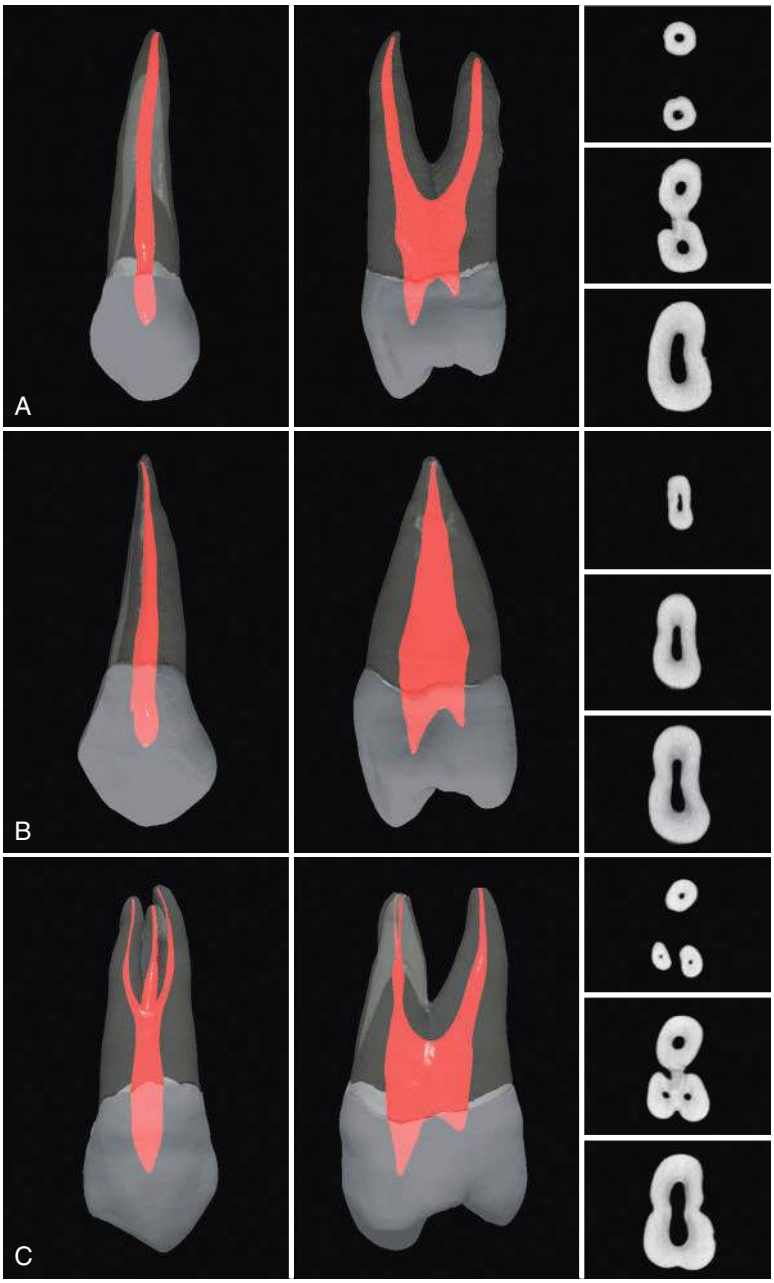


Fig. 7.55 μCT scans of maxillary first premolars. **A,** Common anatomic presentation of this tooth showing two roots. **B,** Premolar with only one canal. **C,** Premolar with three roots. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

three canals can occur in a single root.[32] The scans for the maxillary second premolar can be seen in Fig. 7.57.

MAXILLARY FIRST MOLAR

The maxillary first molar is the largest tooth in volume and one of the most complex in root and canal anatomy (Fig. 7.58).[25] The pulp chamber is widest in the buccolingual dimension, and four pulp horns are present (mesiobuccal, mesiopalatal, distobuccal, and distopalatal). The pulp chamber's cervical outline form has a rhomboid shape, sometimes with rounded corners. The mesiobuccal angle is an acute angle; the distobuccal angle is an obtuse angle; and the palatal angles are basically right angles. The palatal canal orifice is centered palatally; the distobuccal orifice is near the obtuse angle of the pulp chamber floor; and the main mesiobuccal canal orifice is buccal and mesial to the distobuccal orifice and is positioned within the acute

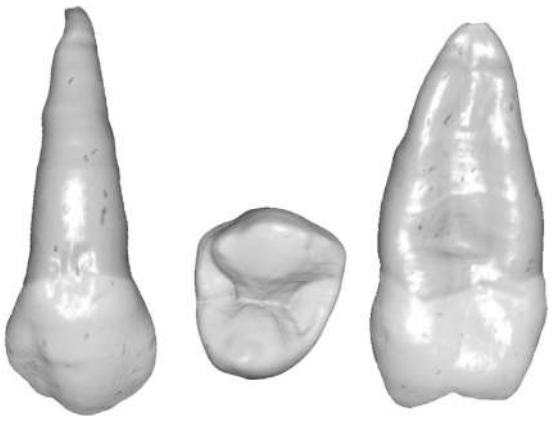


Fig. 7.56 Maxillary second premolar. Average time of eruption—10 to 12 years; average age of calcification—12 to 14 years; average length—21.5 mm. Root curvature (most common to least common): distal, bayonet, buccal, straight.

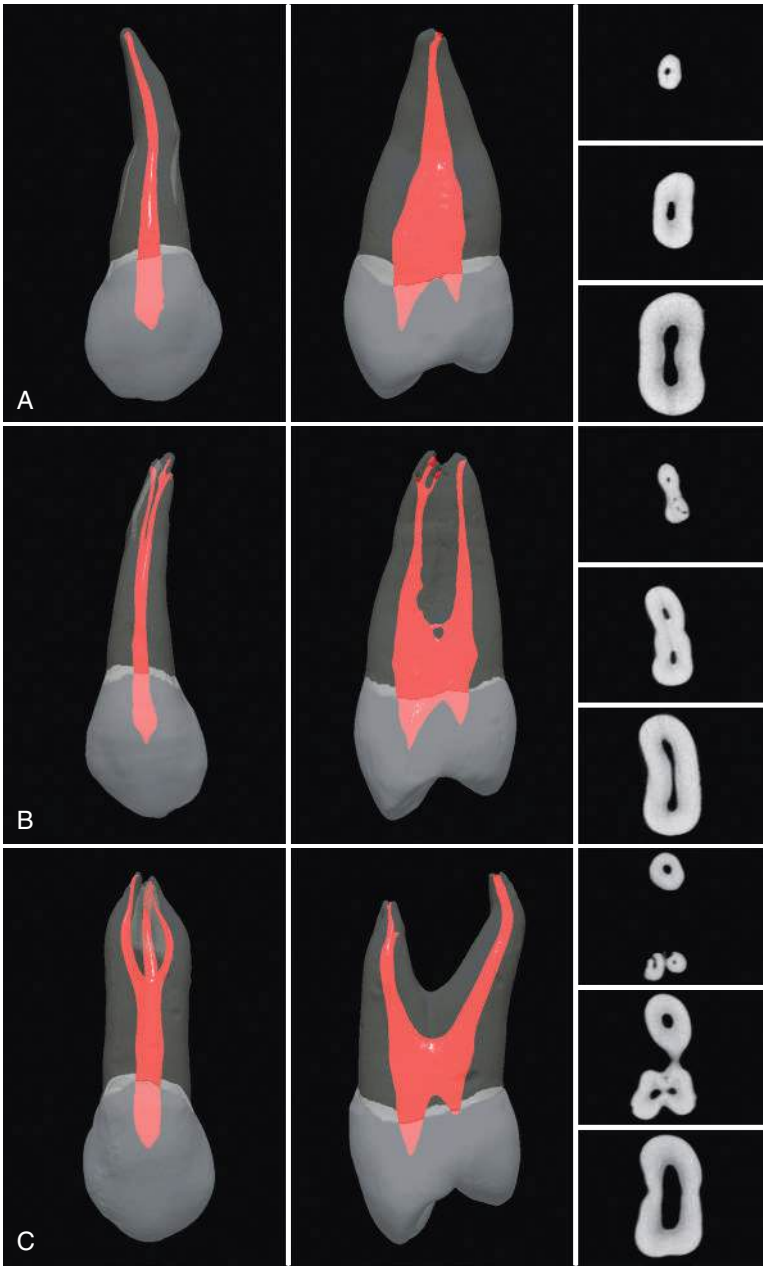


Fig. 7.57 μCT scans of maxillary second premolars. **A,** Common anatomic presentation showing one canal. **B,** Second premolar with two canals and an apical delta. **C,** Second premolar with three roots/canals that divide at the junction of the middle and apical third of the main root. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

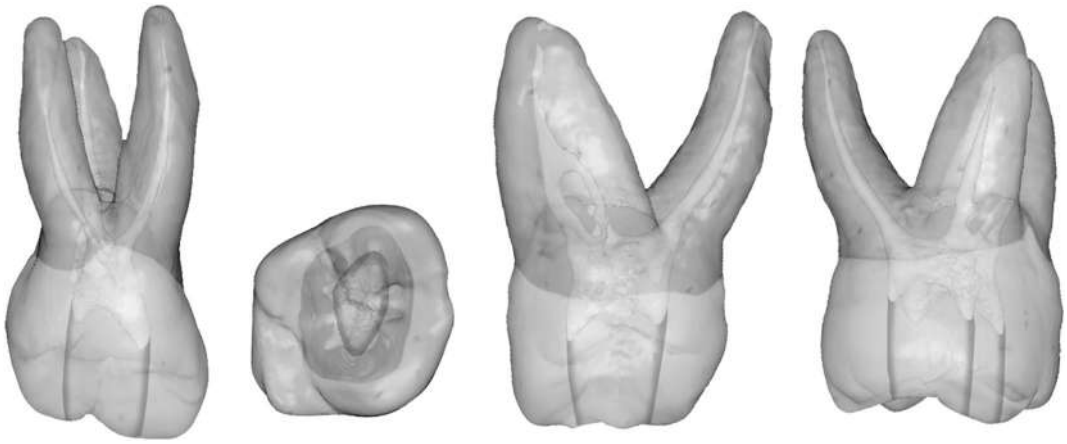


Fig. 7.58 Maxillary first molar. Average time of eruption—6 to 7 years; average age of calcification—9 to 10 years; average length—20.8 mm. Root curvature (most common to least common): mesiobuccal root—distal, straight; distobuccal root—straight, mesial, distal; palatal root—buccal, straight.

angle of the pulp chamber. When there is only one canal in mesiobuccal root, the orifice can be round, oval, or flat. The mesiopalatal canal orifice (also referred to as the *MB-2*) is located palatal and mesial to the mesiobuccal orifice. It can be located close to or far away from mesiobuccal orifice. A line drawn to connect the three main canal orifices—the mesiobuccal orifice, distobuccal orifice, and palatal orifice—forms a triangle, known as the *molar triangle.* The palatal root often curves buccally at the apical one third, which may not be obvious on a standard periapical radiograph. The μCT scans for the maxillary first molar can be seen in Fig. 7.59.

MAXILLARY SECOND MOLAR

Coronally, the maxillary second molar may closely resemble the maxillary first molar (Fig. 7.60). The root and canal anatomy are similar to those of the first molar, although there are differences. The distinguishing morphologic feature of the maxillary second molar is that its three roots are generally grouped closer together and are sometimes fused The μCT scans for the maxillary second molar can be seen in Fig. 7.61.

MAXILLARY THIRD MOLAR

Loss of the maxillary first and second molars is often the reason the third molar must be considered a strategic abutment (Fig. 7.62). However, sometimes a third molar is intentionally transplanted to a vacated first molar position; the same being true for mandibular third molars.[48] The radicular anatomy of the third molar is completely unpredictable, and it may be advisable to explore the root canal morphology to evaluate the likelihood and degree of success. This tooth can have one to four roots and one to six canals, and C-shaped canals also can occur. The third molar usually has three roots and three root canals. The tooth may be tipped significantly to the distal, the buccal, or both, which creates an even greater access problem than with the second molar. The μCT scans for the maxillary molar can be seen in Fig. 7.63.

MANDIBULAR CENTRAL AND LATERAL INCISORS

The root canal systems and access cavities for the two mandibular incisors are so similar they are discussed together (Fig. 7.64). As with the maxillary incisors, a lingual shoulder must be eliminated to allow direct-line access. The shoulder conceals the orifice to a second canal that, if present, is found immediately beneath it. Most mandibular incisors have a single root with what radiographically appears to be a long, narrow canal. However, it is a broad canal labiolingually. Often a dentinal bridge is present in the pulp chamber that divides the root into two canals. The two canals usually join and exit through a single AF, but they may persist as two separate canals (Figs. 7.65 and 7.66). Occasionally, one canal branches into two canals, which subsequently rejoin into a single canal before reaching the apex.

MANDIBULAR CANINE

The root canal system of the mandibular canine is very similar to that of the maxillary canine, except that the dimensions are smaller, the root and root canal outlines are narrower in the mesiodistal dimension, and the mandibular canine occasionally has two roots and two root canals located labially and lingually (Fig. 7.67). The root canal of the mandibular cuspid is narrow mesiodistally but usually very broad buccolingually, and the access cavity for the mandibular canine is oval or slot-shaped.[112] The μCT scans of the mandibular canine can be seen in Fig. 7.68.

MANDIBULAR FIRST PREMOLAR

As a group, the mandibular premolars present significant anatomic challenges because of the extreme variations in their root canal morphology (Fig. 7.69).[63,82,121] The root canal system of the mandibular first premolar is wider buccolingually than mesiodistally.[123] Two pulp horns are present: a large, pointed buccal horn and a small, rounded lingual horn. At the cervical line, the root and canal are oval; this shape tends to become round as the canal approaches

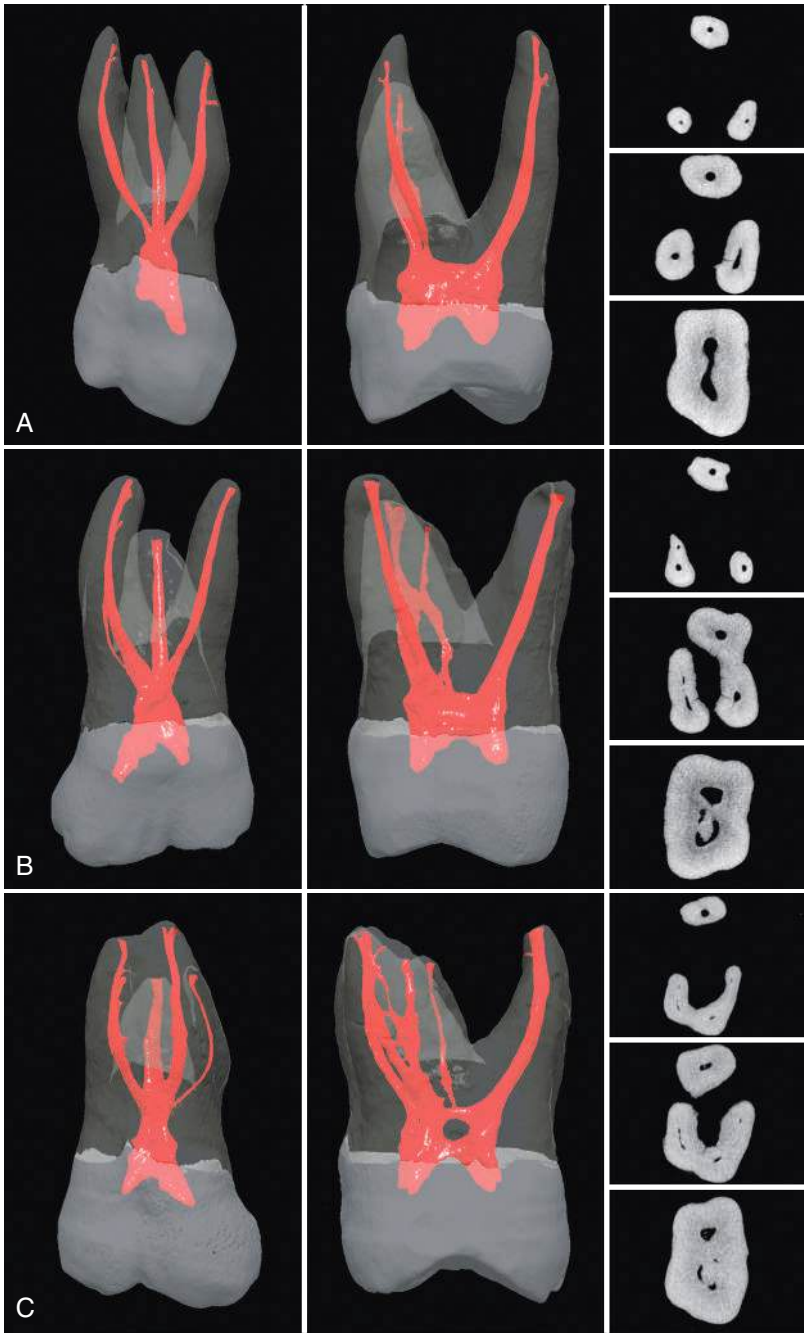


Fig. 7.59 μCT scans of maxillary first molars. **A,** Common anatomic presentation showing accessory/lateral canals. **B,** First molar with four canals, with mesiobuccal and mesiopalatal sharing an anastomosis in the mid-root. **C,** Maxillary molar with four pulp horns, five canals, and significant anastomoses between the canals. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

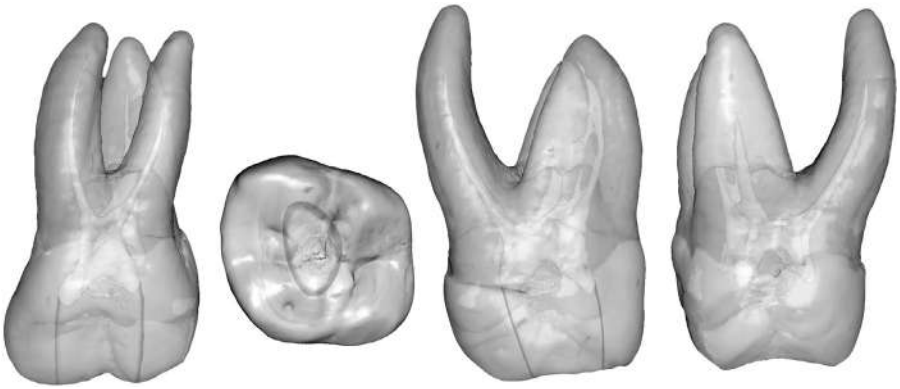


Fig. 7.60 Maxillary second molar. Average time of eruption—11 to 13 years; average age of calcification—14 to 16 years; average length—20 mm. Root curvature (most common to least common): mesiobuccal root—distal, straight; distobuccal root—straight, mesial, distal; palatal root—straight, buccal.

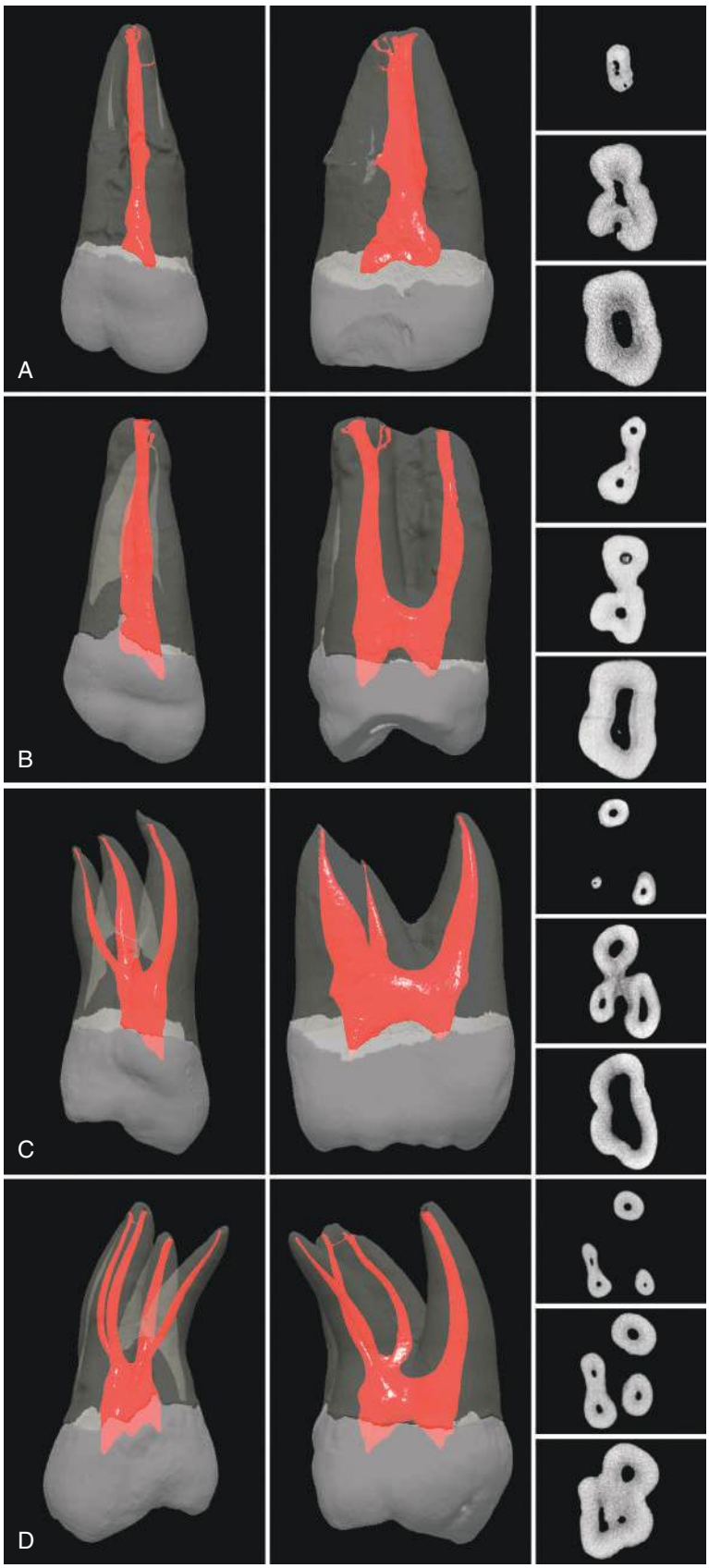


Fig. 7.61 μCT scans of maxillary second molars; four possible variations. **A,** Uncommon anatomic presentation of this tooth with one canal. **B,** Second molar with two canals. **C,** Second molar with three canals. **D,** Second molar with four distinct canals. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

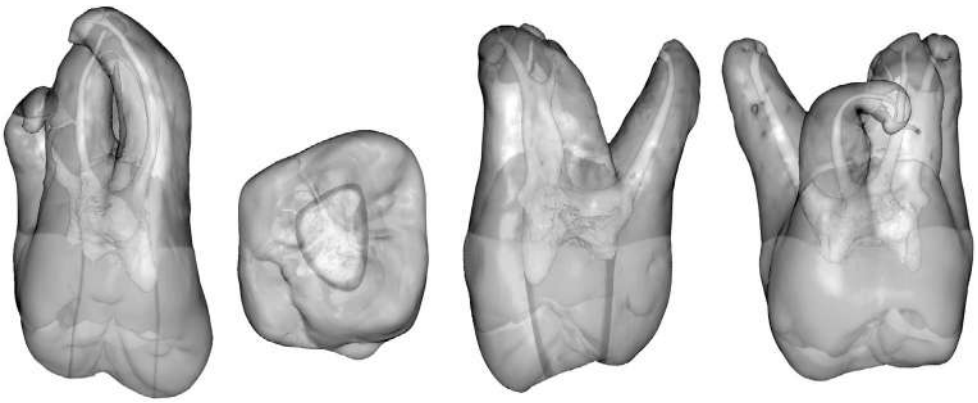


Fig. 7.62 Maxillary third molar. Average time of eruption—17 to 22 years; average age of calcification—18 to 25 years; average length—17 mm.

the middle of the root. If two canals are present, they tend to be round from the pulp chamber to their foramen. Direct access to the buccal canal is usually possible, whereas the lingual canal may be quite difficult to find. To address this situation, the lingual wall of the access cavity may have to be extended farther lingually; this makes the lingual canal easier to locate. The mandibular first premolar sometimes may have three roots and three canals. Multiple studies have reported C-shaped canal anatomy in this tooth.[10,29,31,68,80] The μCT scans for the mandibular first premolar can be seen in Fig. 7.70.

MANDIBULAR SECOND PREMOLAR

The mandibular second premolar is similar to the first premolar, with the following differences: The lingual pulp horn usually is larger, the root and root canal are more often oval than round, the pulp chamber is wider buccolingually, and the separation of the pulp chamber and root canal is normally distinguishable compared with the more regular taper in the first premolar (Fig. 7.71). The canal morphology of the mandibular second premolar is similar to that of the first premolar with its many variations: two, three, and four canals and a lingually tipped crown. Fortunately, these variations are found less often in the second premolar. The μCT scans for the mandibular second premolar can be seen in Fig. 7.72.

MANDIBULAR FIRST MOLAR

The mandibular first molar seems to be the tooth that most often requires an endodontic procedure (vital pulp capping, pulpotomy, root canal therapy); therefore, its morphology has received a great deal of attention (Fig. 7.73).[125] The occlusal appearance reflects a Y5 groove and cuspal pattern and the tooth usually has two roots, but occasionally it has three, with two or three canals in the mesial root and one, two, or three canals in the distal root. The μCT scans of the mandibular first molar can be seen in Fig. 7.74.

The canals in the mesial root are the mesiobuccal and mesiolingual canals. A middle mesial canal (MMC) sometimes is present in the developmental groove between the other mesial canals,[4,12,76,79,104,107,111] but it may only represent a wide anastomosis between the two mesial canals (Fig. 7.75), or in some cases it may actually be a separate canal from orifice to apex.[89] Troughing into the groove between the mesiobuccal and mesiolingual orifices may be indicated to identify this canal,[9,43,44] if present, or it may just result in blind search within the anastomosis with the risk of a perforation or root weakening due to the commonly found external invagination found both mesially and distally. The presence of two separate distal roots is rare, but this does occur. The mesial canal orifices are usually well separated in the main pulp chamber and connected by a developmental groove.[19]

When only one distal canal is present, the orifice is oval buccolingually and the opening generally is located distal to the buccal groove. This orifice usually can be explored from the mesial with either a DG-16 explorer or a small K-file. If the file tip takes a sharp turn in a distobuccal or distolingual direction, the clinician should search for yet another orifice.

Multiple accessory foramina may be located in the furcation of the mandibular molars.[37] These foramina are usually impossible to clean and shape directly, and therefore, extensive irrigation protocols should be applied.

The traditional access cavity for the mandibular first molar typically is trapezoid or rhomboid, regardless of the number of canals present.[120] When four or more canals are present, the corners of the trapezoid or rhombus should correspond to the positions of the main orifices. A variation in root morphology is the presence of an extra distolingual root.[103] The radix entomolaris (RE) is a supernumerary root located distolingually in mandibular molars (Fig. 7.76),[7,16] whereas the radix paramolaris (RP) is an extra root located mesiobuccally.[102]

MANDIBULAR SECOND MOLAR

The mandibular second molar is somewhat smaller coronally than the first molar and tends to be more symmetric (Fig. 7.77). This tooth is identified by the proximity of its roots and its +4 occlusal groove and cuspal pattern. The two roots often sweep distally in a gradual curve, with the apices close together. In some teeth only one root is present. This tooth may have one, two, three, or four root canals. The μCT scans for the mandibular second molar can be seen in Fig. 7.78. The access cavity for a two-canal second molar is rectangular, wide mesiodistally and narrow buccolingually. The access cavity for a single-canal mandibular

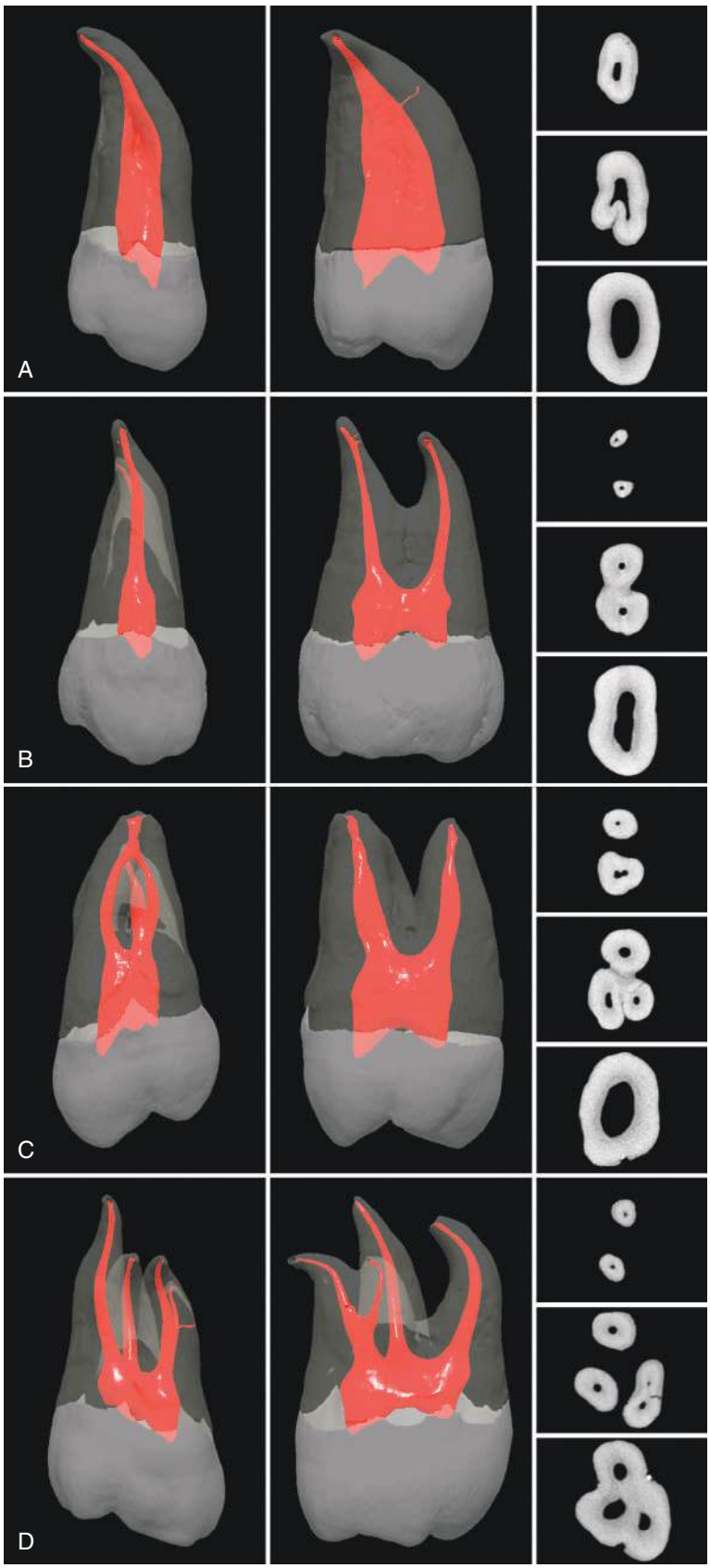


Fig. 7.63 μCT scans of maxillary third molars show a range of anatomic variations. **A,** Single-canal tooth. **B,** Two-rooted third molar. **C,** Two-rooted, three-canal third molar with significant root curvatures. **D,** Three-rooted, four-canal third molar. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

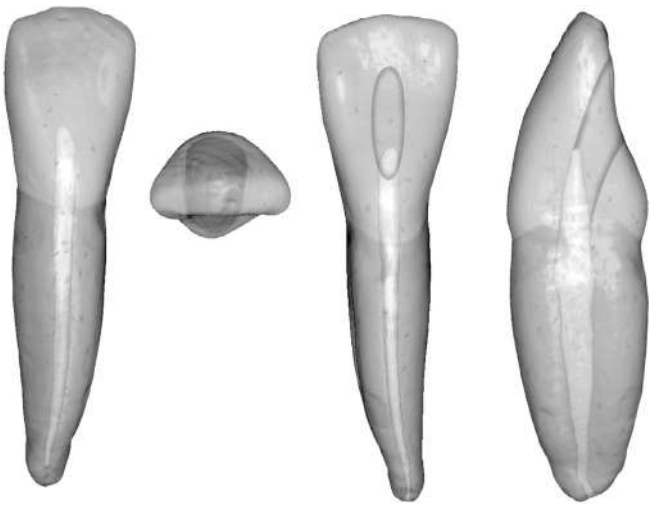


Fig. 7.64 Mandibular central/lateral incisors. Average time of eruption—6 to 8 years; average age of calcification—9 to 10 years; average length—20.7 mm. Root curvature (most common to least common): straight, distal, labial.

second molar is oval and is lined up in the center of the occlusal surface.

MANDIBULAR THIRD MOLAR

The mandibular third molar is anatomically unpredictable and must be evaluated on the basis of its root formation (Fig. 7.79).[52] More often than not it has an occlusal pattern similar to the first molar only much smaller in overall size. Fused short, severely curved, or malformed roots often support well-formed crowns. This tooth may have one to four roots and one to six canals. The μCT scans for the mandibular third molar can be seen in Fig. 7.80.

C-shaped canals can also occur (Fig. 7.81).[40] Many of these teeth can be successfully root treated, regardless of anatomic irregularities; however, the long-term prognosis is determined by the root surface volume in contact with bone. The clinician must weigh the benefit of treatment against the prognosis.

TEETH WITH C-SHAPED ROOT CANAL SYSTEMS

The main cause for C-shaped roots and canals is the failure of Hertwig's epithelial root sheath to fuse on either the buccal or lingual root surface (Fig. 7.82).[40] The C-shaped canal system can assume many variations in its morphology (Fig. 7.83). Further cross section of types I, II, and III are seen in Fig. 7.83. The original classification[67] has been modified and has produced a more detailed description of C-shaped root and canal morphology. The C-shaped canal configuration can vary along the root depth so that the appearance of the orifices may not be good predictors of the actual canal anatomy.[26,28,70,117,126]

Most C-shaped canals occur in the mandibular second molar (Fig. 7.84),[117] but they also have been reported in the mandibular first molar,[118] the maxillary first and second molars, and the mandibular first premolar.[26,27,29,30,75,78,116,121,123,125,126]

The access cavity for teeth with a C-shaped root canal system varies considerably and depends on the pulp morphology of the specific tooth. Teeth with C-shaped anatomy pose a considerable technical challenge; therefore, use of the DOM during all treatment phases is recommended.[15]

The 3D reconstructed images in this chapter were obtained from the tooth and canal morphology database at the School of Stomatology, Wuhan University, China. The database was established by Dr. Bing Fan's group and supported by the National Natural Science Foundation of China (grant no. 30572042, 30872881, 81070821) and the Key Technologies R&D Programme of Hubei Province of China (grant no. 2007AA302B06). The microcomputed tomography machine used for scanning was the μCT-50, Scanco Medical, Bassersdorf, Switzerland. The multiple software used for 3D reconstruction included 3D-Doctor (Able Software Corp., Lexington, MA) and VGStudio MAX (Volume Graphics GmbH, Heidelberg, Germany).

Access cavity illustrations: Designed and formatted by Dr. Richard Burns, San Mateo, CA; and Dr. Eric Herbranson, San Leandro, CA.

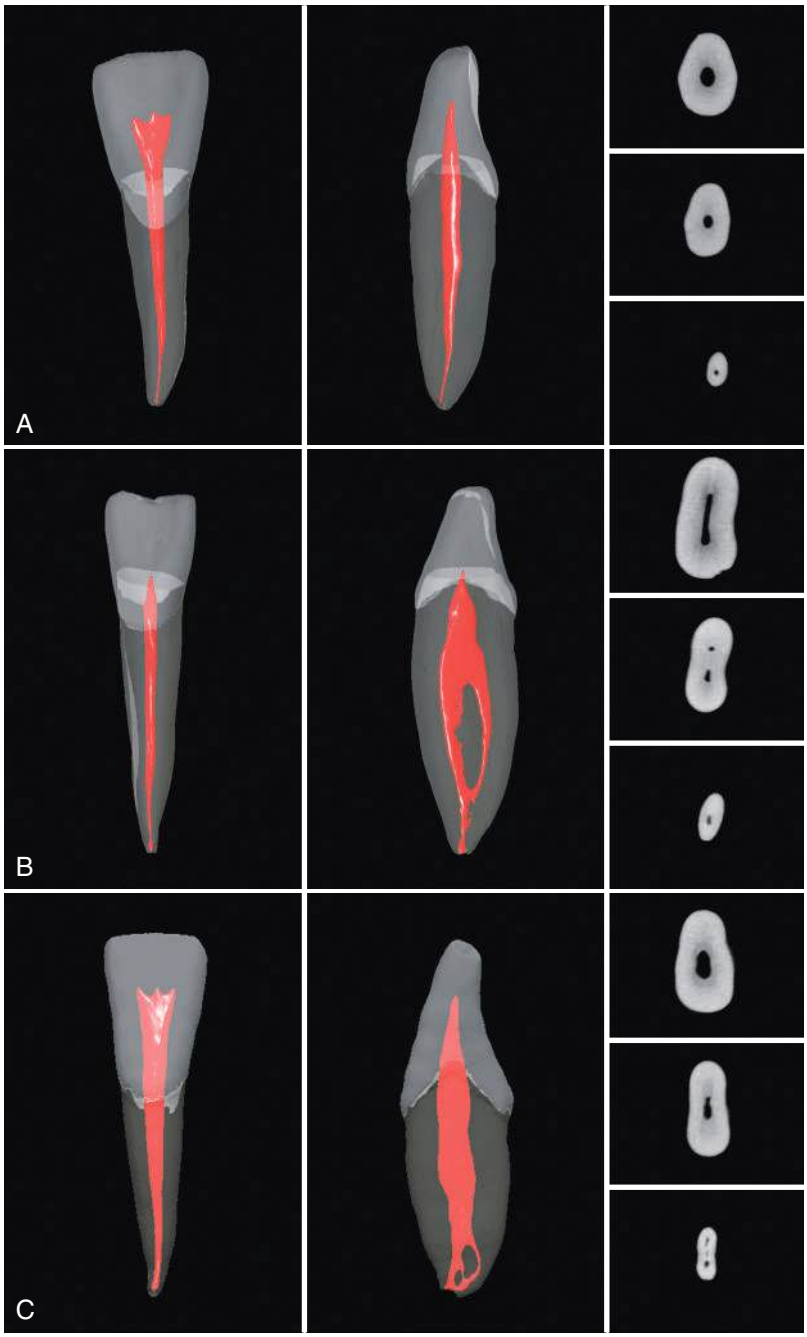


Fig. 7.65 μCT scans of mandibular central incisors. **A,** Common anatomic presentation. **B,** Central incisor with two canals. **C,** Central incisor with an apical delta. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

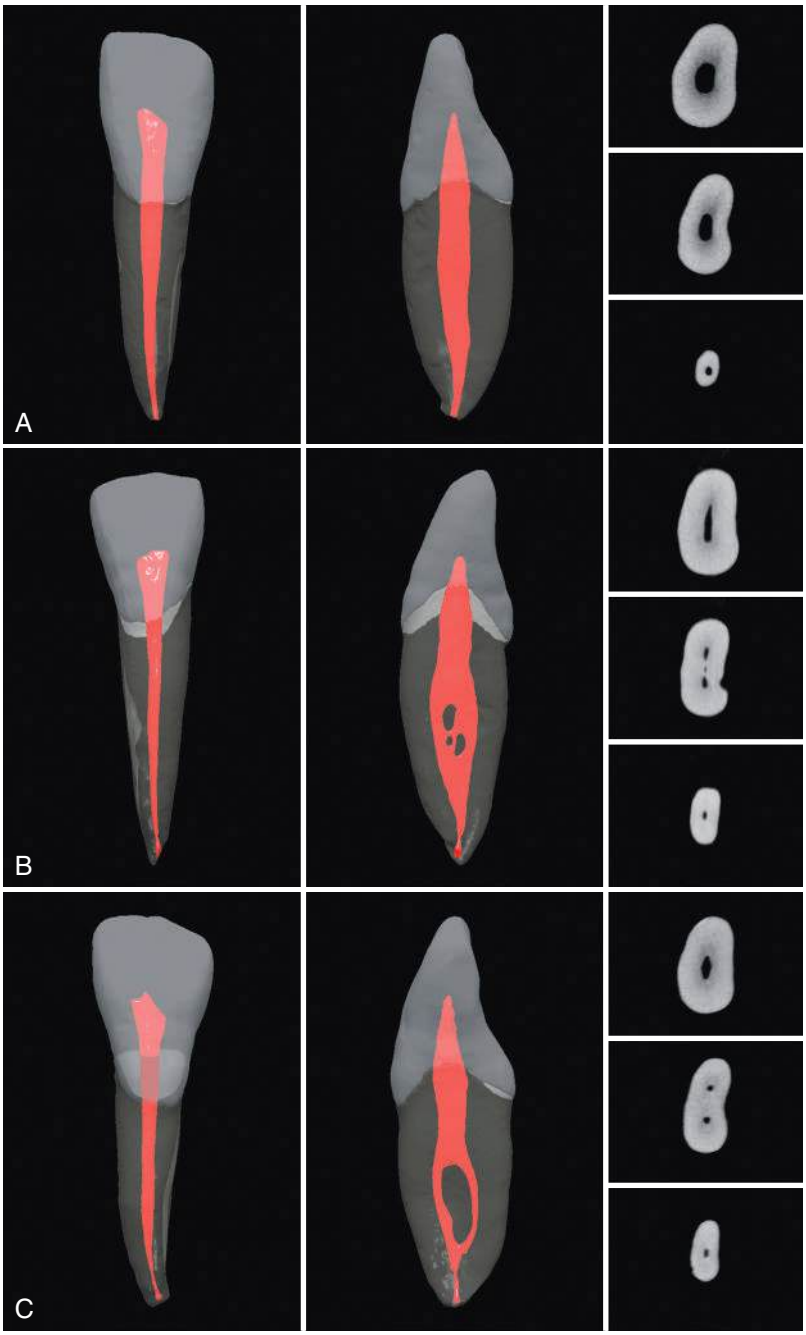


Fig. 7.66 μCT scans of mandibular lateral incisors. **A,** Common anatomic presentation. **B,** Lateral incisor with broad, thin buccolingual anatomy. **C,** Lateral incisor in which the canal splits into two but returns to form one canal apically. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

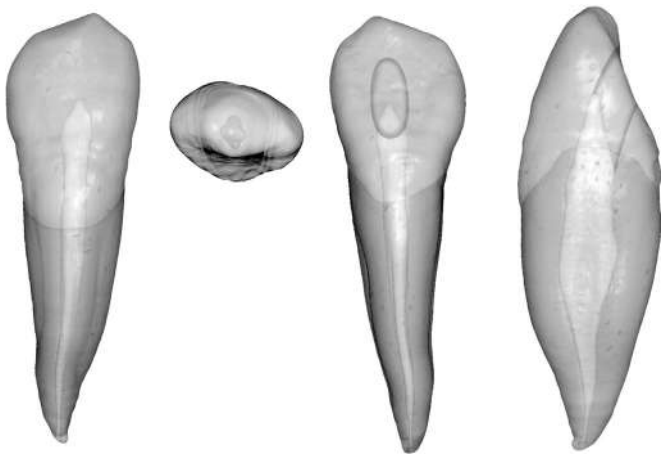


Fig. 7.67 Mandibular canine. Average time of eruption—9 to 10 years; average age of calcification—13 years; average length—25.6 mm. Root curvature (most common to least common): straight, distal, labial.

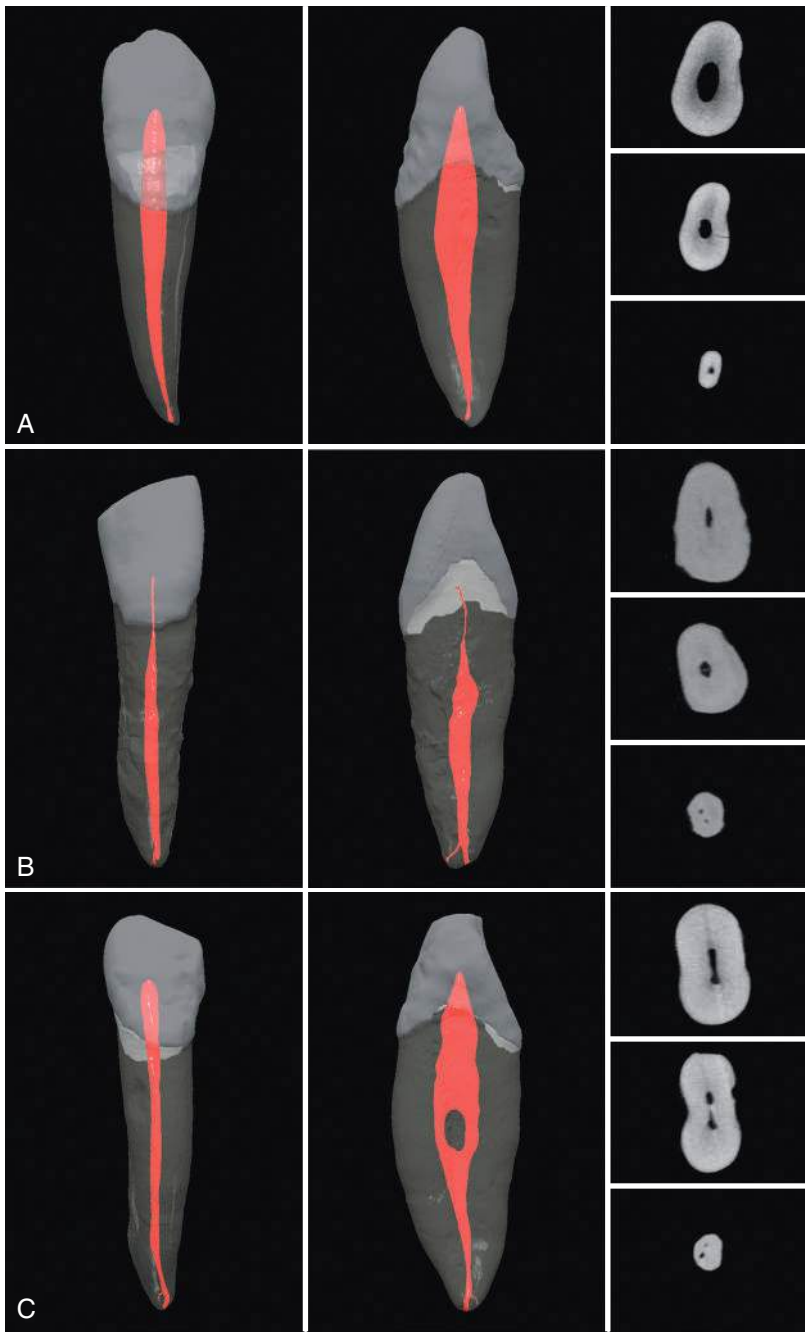


Fig. 7.68 μCT scans of mandibular canines. **A,** Common anatomic presentation. **B,** Canine with an extra apical canal. **C,** Canine that splits into two but returns to one canal apically. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

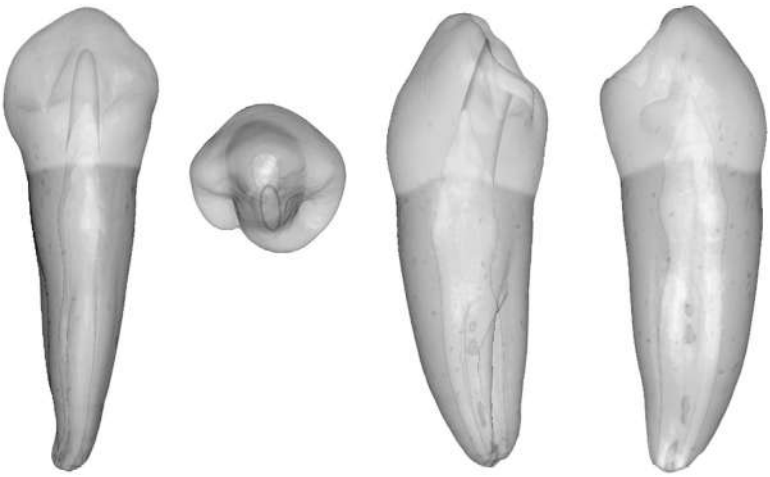


Fig. 7.69 Mandibular first premolar. Average time of eruption—10 to 12 years; average age of calcification—12 to 13 years; average length—21.6 mm. Root curvature (most common to least common): straight, distal, buccal.

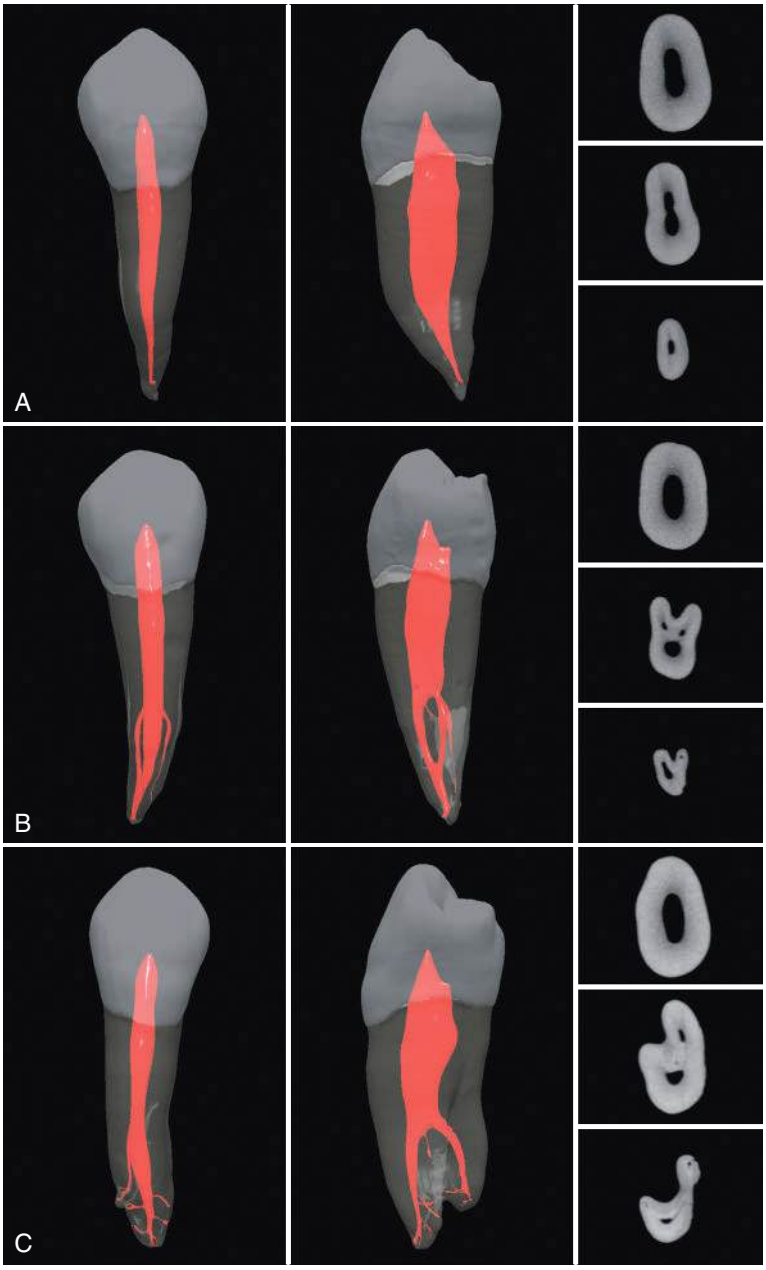


Fig. 7.70 μCT scans of mandibular first premolars. **A,** Common anatomic presentation. **B,** First premolar with significant canal deviations in the middle to apical third before returning to a single large canal apically and a small deviating canal to the proximal. **C,** First premolar with a branching main canal lingually and multiple accessory canals. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

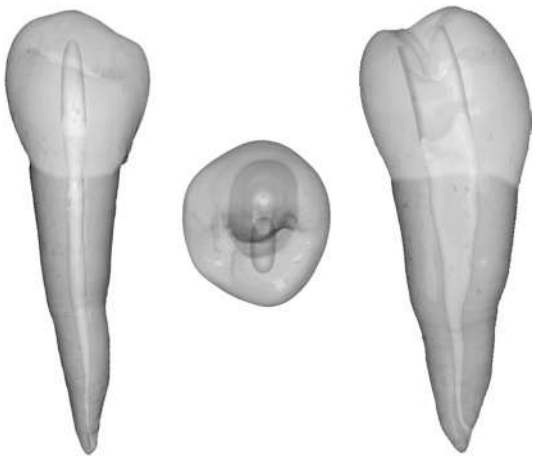


Fig. 7.71 Mandibular second premolar. Average time of eruption—11 to 12 years; average age of calcification—13 to 14 years; average length—22.3 mm. Root curvature (most common to least common): straight, distal, buccal.

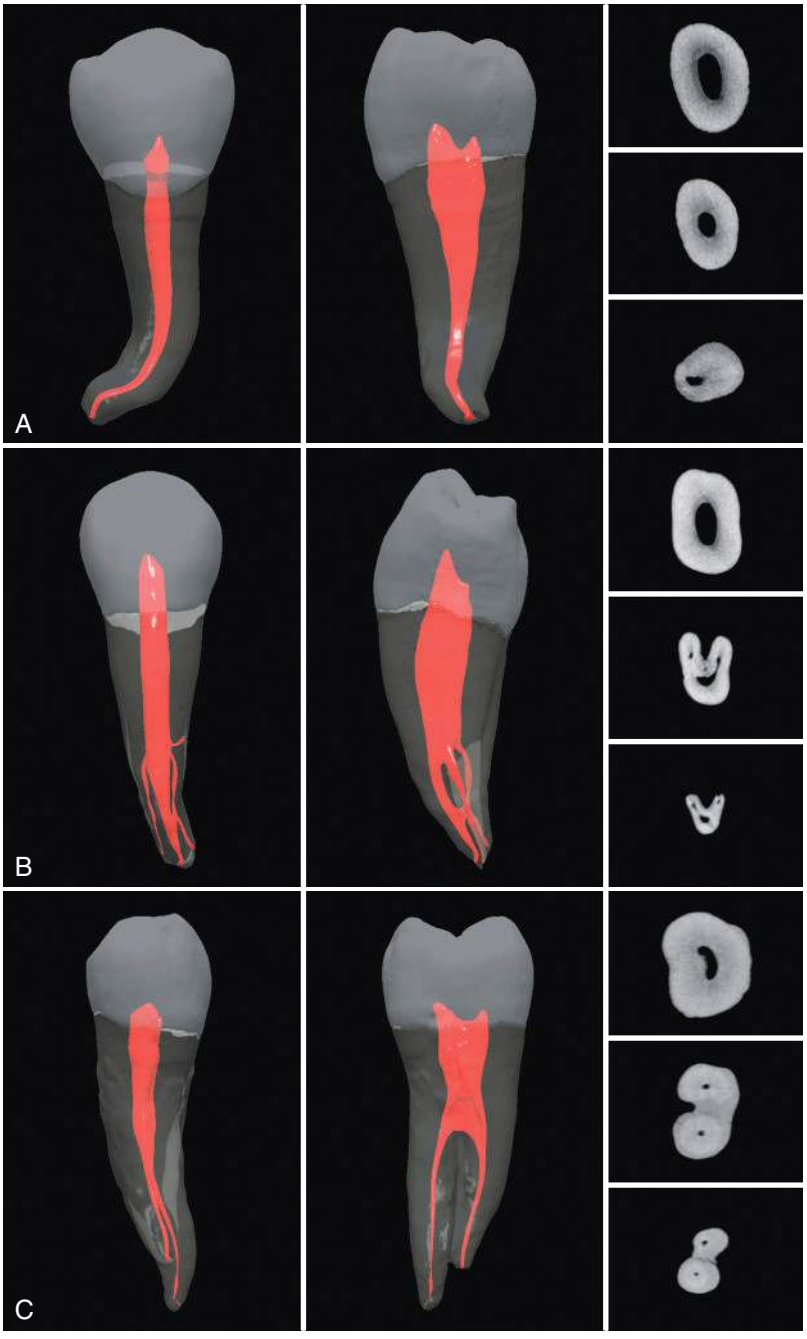


Fig. 7.72 μCT scans of mandibular second premolars. **A,** Common anatomic presentation. **B,** Second premolar with significant canal deviations in the middle to apical third. **C,** Second premolar with fused root that exhibits two distinct canals. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

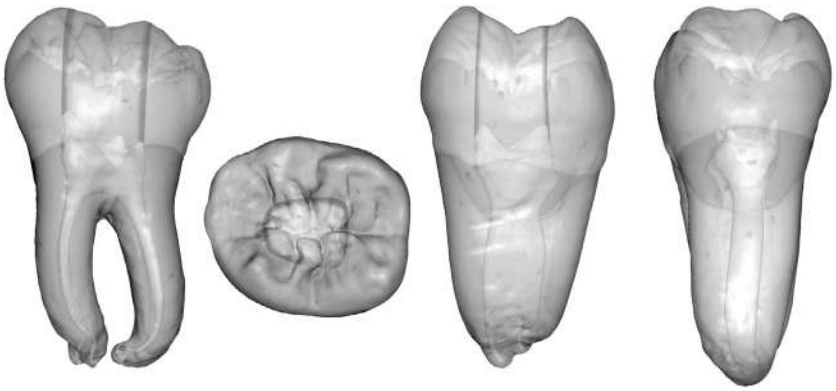


Fig. 7.73 Mandibular first molar. Average time of eruption—6 years; average age of calcification—9 to 10 years; average length—21 mm. Root curvature (most common to least common): mesial root—distal, straight; distal root—straight, distal.

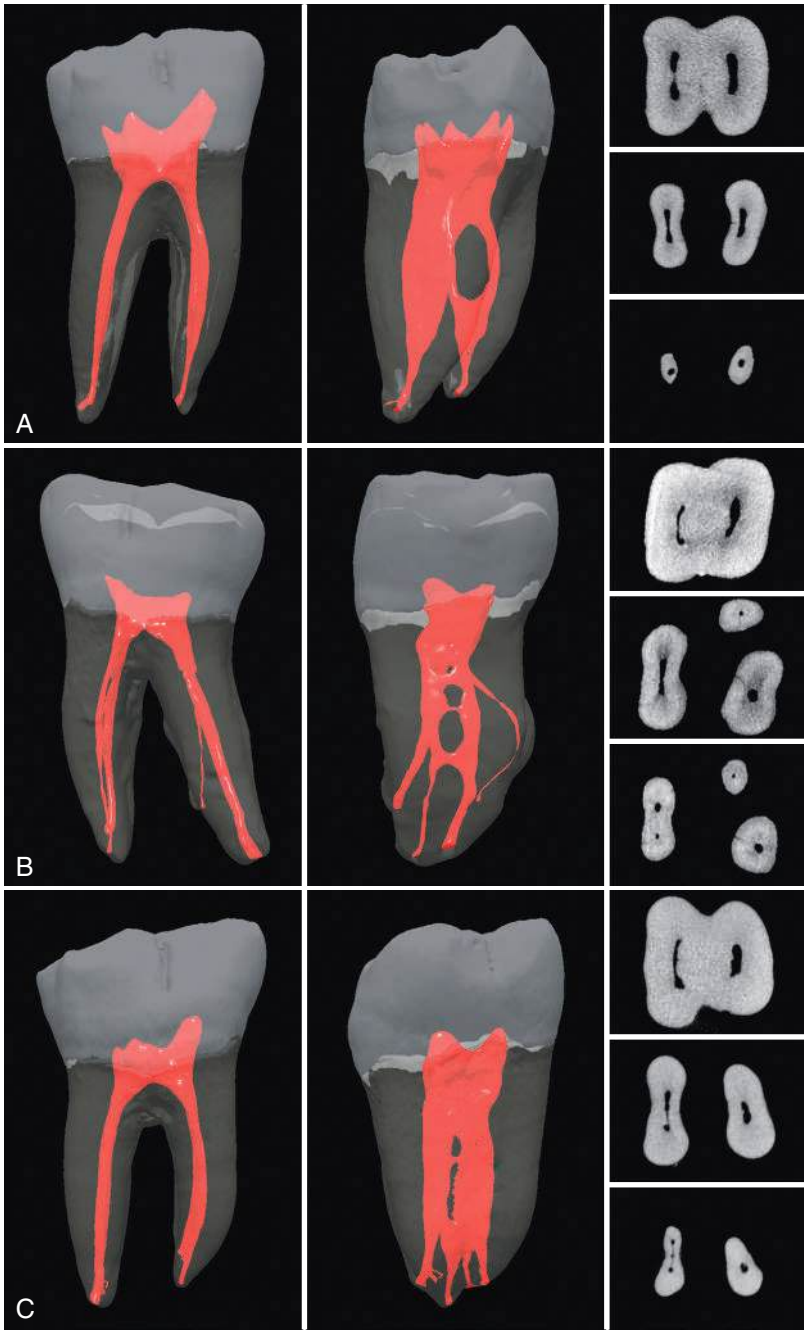


Fig. 7.74 μCT scans of mandibular first molars. **A,** Common anatomic presentation. **B,** First molar with three main canals and a deviant fourth canal/fourth root. **C,** First molar with wide connections or anastomoses between the mesial canals, demonstrating multiple canal exits. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

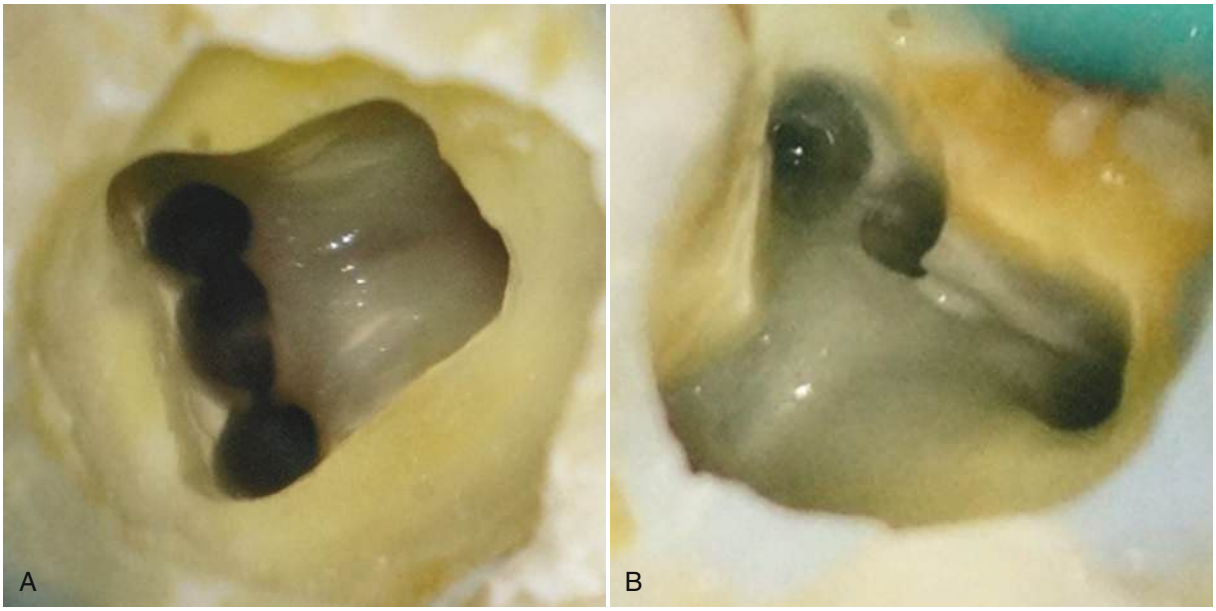


Fig. 7.75 A, A photograph (×8) of a mandibular molar showing fins between MB and ML canals**. B,** A photograph (×8) of a mandibular molar showing three mesial canals. *MB,* Mesiobuccal; *ML,* mesiolingual. (From Nosrat A, Deschenes RJ, Tordik PA, Hicks ML, Fouad AF: Middle mesial canals in mandibular molars: incidence and related factors, *J Endod* 41:28–32, 2015.)

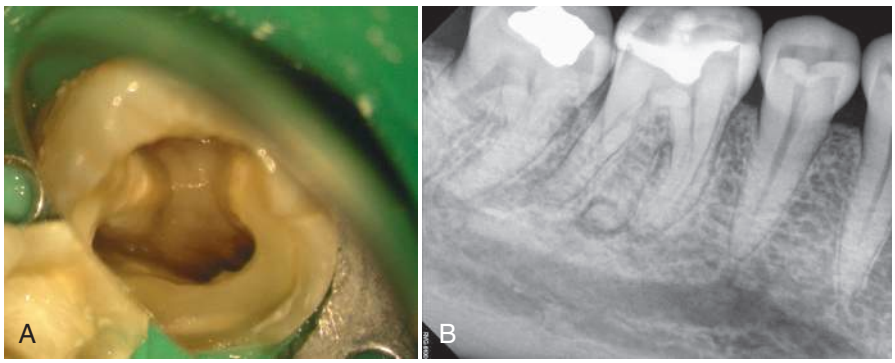


Fig. 7.76 A, Radix entomolaris. Notice lingual position of its orifice in relation to the two canals in the distal root. **B,** Radiograph with obvious radix entomolaris on mandibular first molar that curves to the buccal. The reader is referred to a recent publication that details extensively this type of anatomy (see following credit for part **B**). (**A,** Courtesy Dr. William J. Aippersbach, Venice, FL. **B,** From Abella F, Patel S, Durán-Sindreu F, Mercadé M, Roig M: Mandibular first molars with disto-lingual roots: review and clinical management, *Int Endod J* 45:963–978, 2012.)

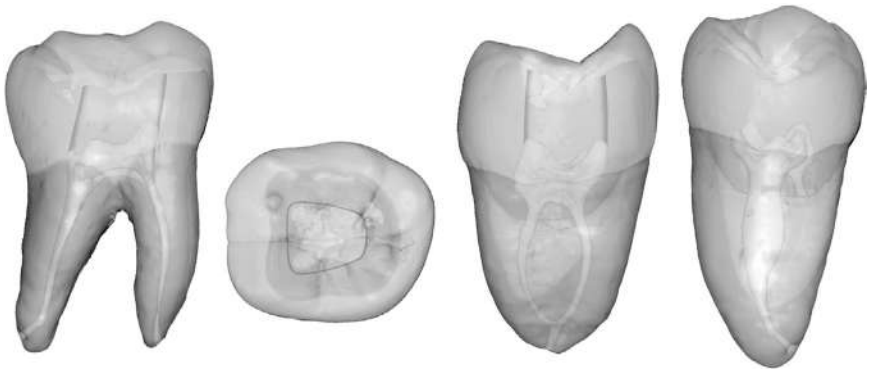


Fig. 7.77 Mandibular second molar. Average time of eruption—11 to 13 years; average age of calcification—14 to 15 years; average length—19.8 mm. Root curvature (most common to least common): mesial root—distal, straight; distal root—straight, distal, mesial, buccal; single root—straight, distal, bayonet, lingual.

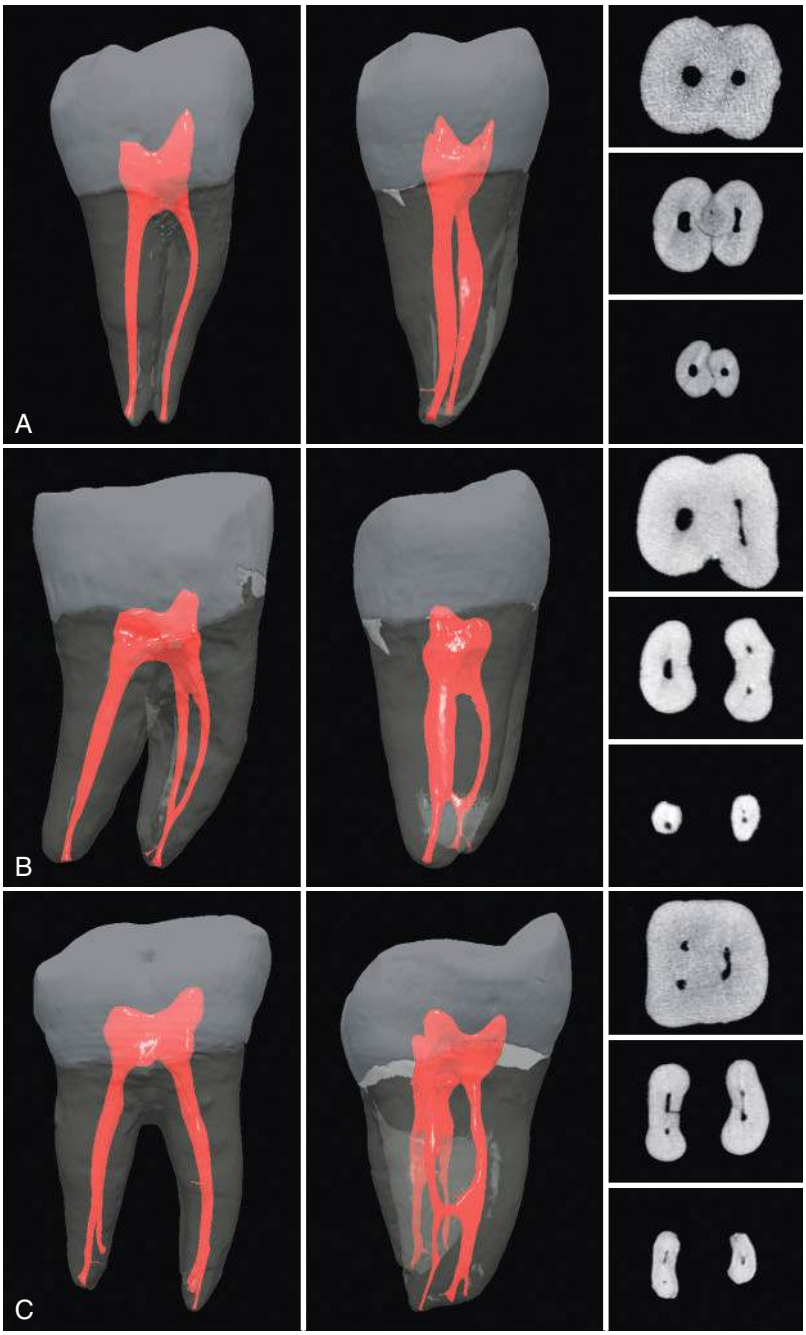


Fig. 7.78 μCT scans of mandibular second molars. **A,** Two-canal second molar with fused roots. **B,** Second molar with three initial canals ending in one canal apically in both roots. **C,** Second molar with four distinct canals. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

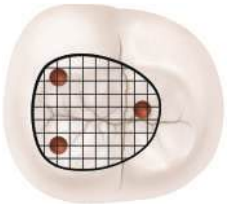


Fig. 7.79 Mandibular third molar. Average time of eruption—17 to 21 years; average age of calcification—18 to 25 years; average length—18.5 mm.

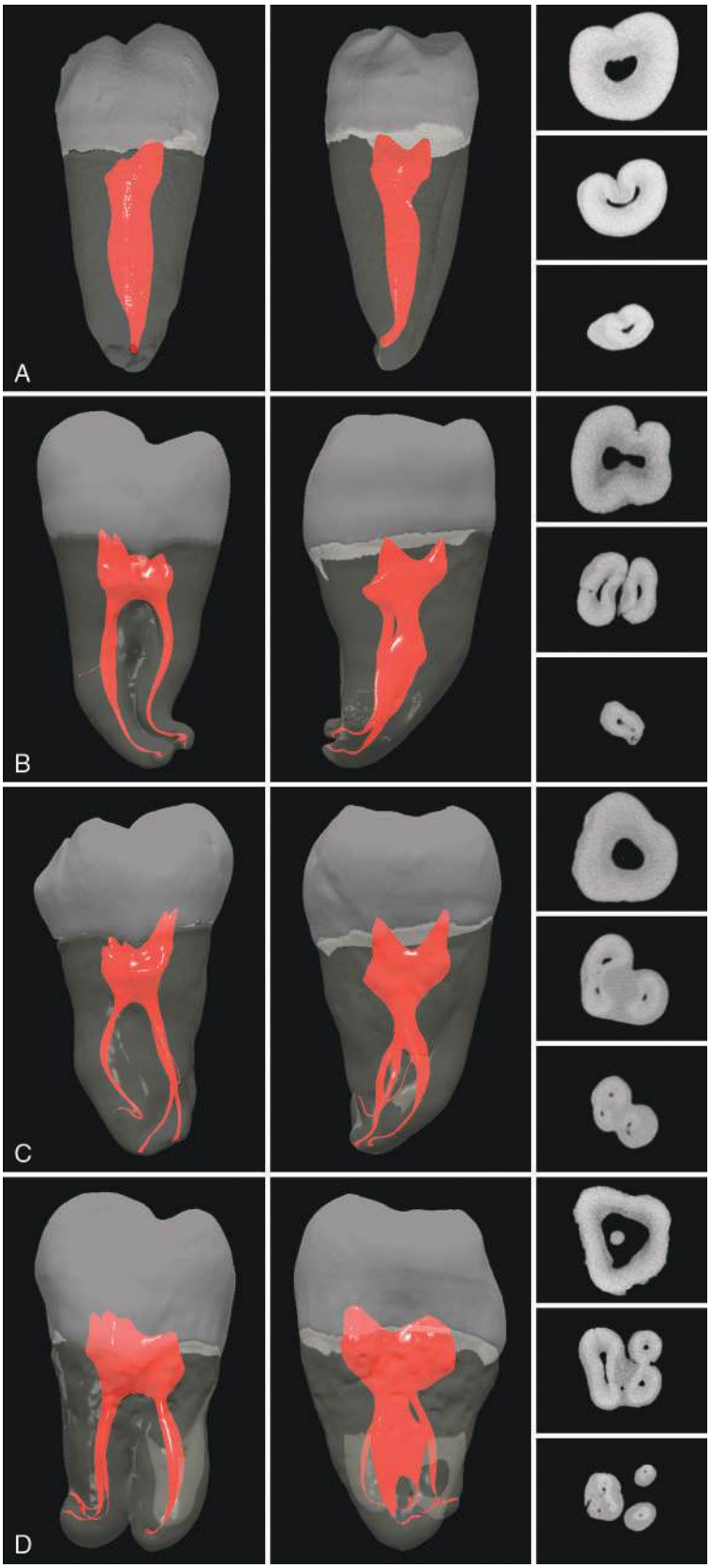


Fig. 7.80 μCT scans of mandibular third molars represent multiple variations for this tooth. **A**, Single C-shaped type canal. **B,** Complex anatomy with significant canal curvatures apically. **C,** Three canals that curve in multiple directions. **D,** Flattened, ribbon-shaped canals with significant apical curvatures. All teeth are shown from both a buccal (vestibular) and a proximal perspective, along with the cross-sectional anatomy at the coronal, middle, and apical levels.

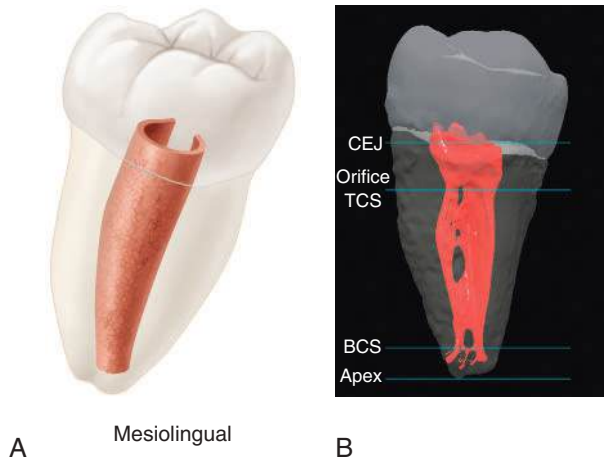


Fig. 7.81 A, Diagram of a C-shaped canal anatomy: one continuous canal from pulp chamber floor to apex. **B,** One large canal moving from the top of the C-shaped canal *(TCS)* to the apical-most or bottom of C-shaped canal *(BCS)*, encompassing most of the canal space. *CEJ,* Cementoenamel junction.

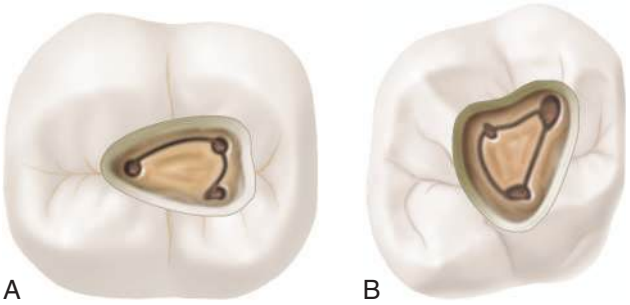


Fig. 7.82 C-shaped canal anatomy. **A,** Mandibular second molar. **B,** Maxillary first molar.

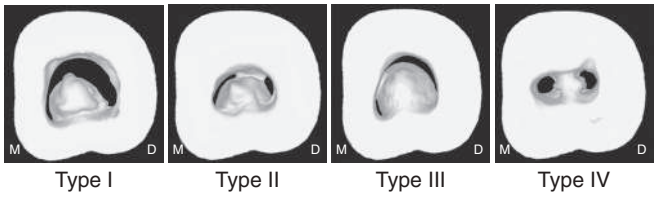


Fig. 7.83 Types of pulpal floor. *M,* Mesial side; *D,* distal side. (From Min Y, Fan B, Cheung G, et al: *C*-shaped canal system in mandibular second molars. III. The morphology of the pulp chamber floor, *J Endod* 32: 1155–1159, 2006.)

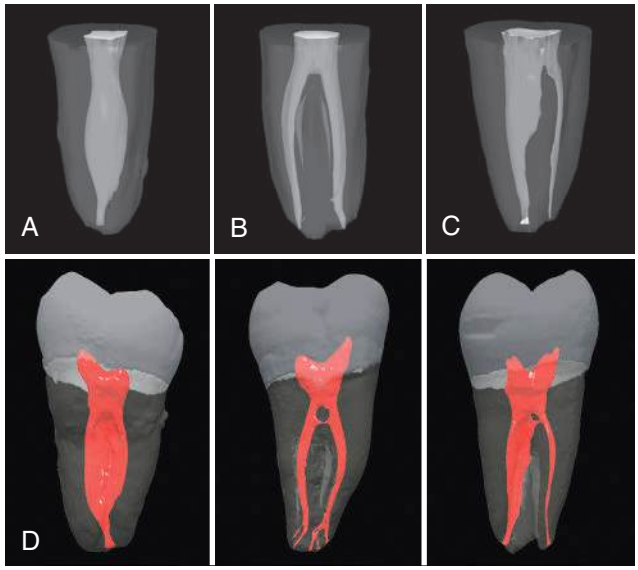


Fig. 7.84 Three-dimensional classification of C-shaped canal configuration. **A,** Merging type. **B,** Symmetrical type. **C,** Asymmetrical type. **D,** Additional C-shaped canal variations. (From Gao Y, Fan B, Cheung G, et al: *C*-shaped canal system in mandibular second molars. IV. Morphological analysis and transverse measurement, *J Endod* 32:1062–1065, 2006.)

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